

Lecture 19: Molecular spectroscopy and photochemistry

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Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

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Major points of last lecture

- **Bonding in transition metal compounds and coordination complexes**
 1. **Chemistry of the Transition Metals**
 2. **Introduction to Coordination Chemistry**
 3. **Structures of Coordination Complexes**
 4. **Crystal Field Theory: Optical and Magnetic Properties**
 5. **Optical Properties and the Spectrochemical Series**
 6. **Bonding in Coordination Complexes**

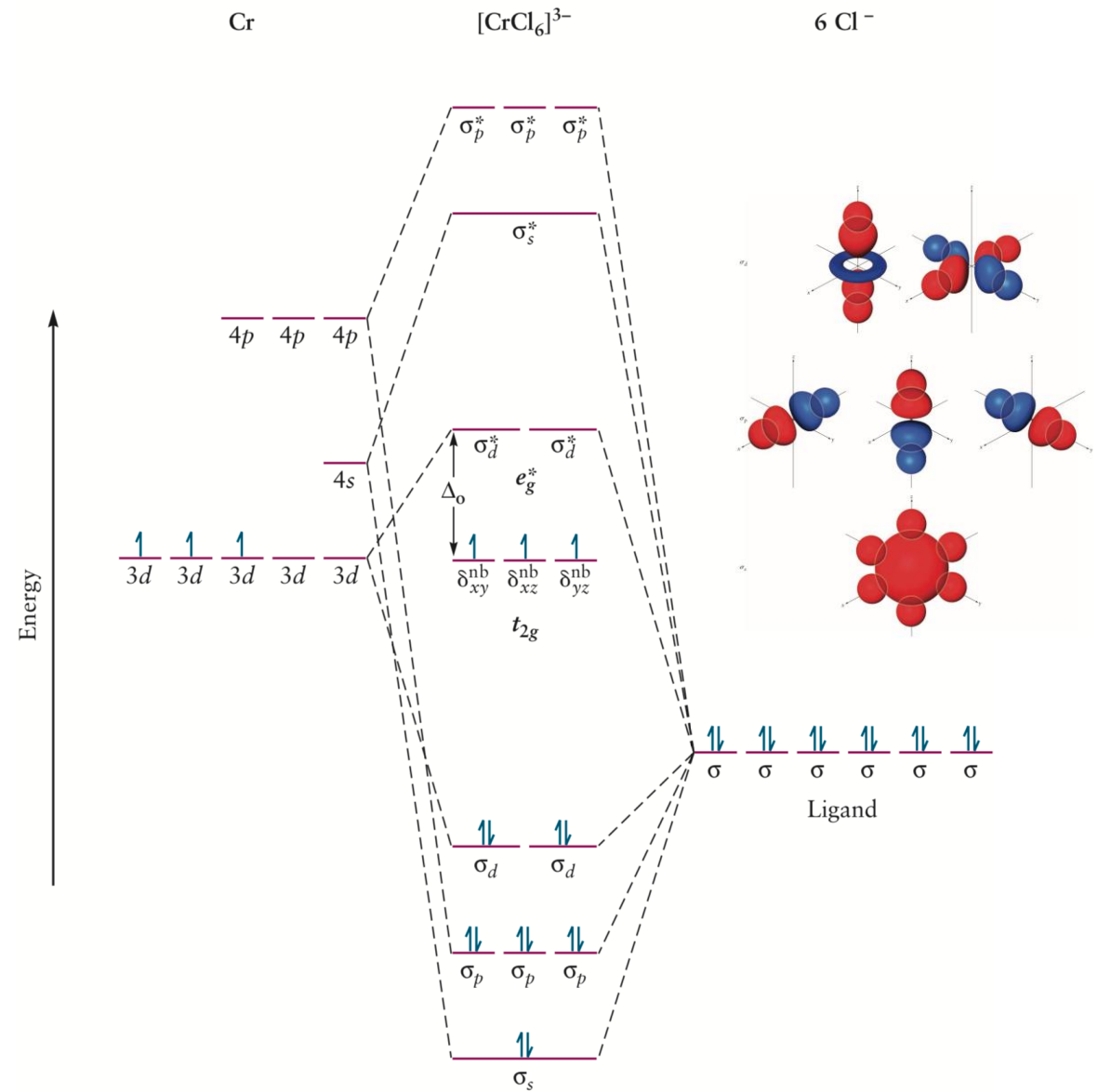
Bonding in Coordination Complexes: Molecular Orbital Theory

Molecular Orbital Theory

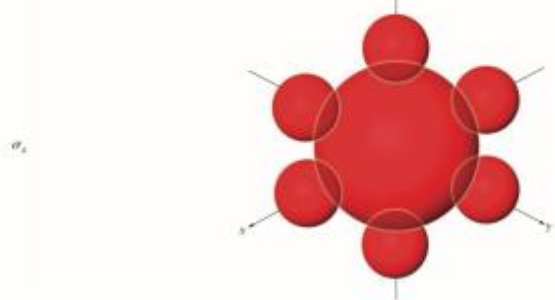
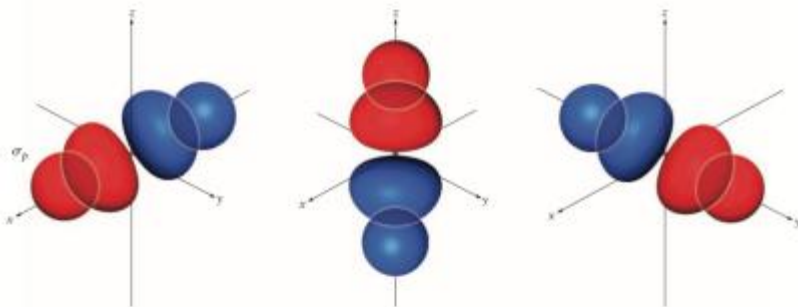
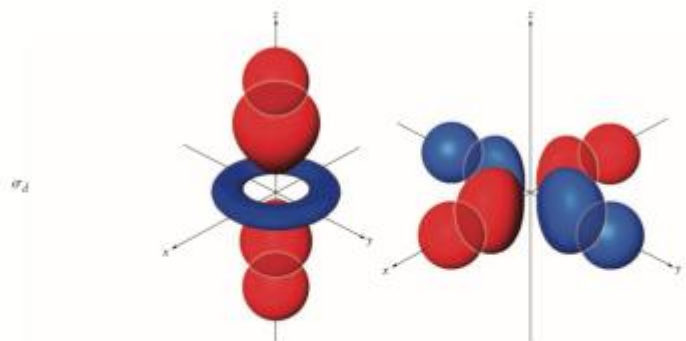
- Metal orbitals: 3d, 4s, and 4p orbitals.
- Ligand orbitals: sp^3 orbitals containing a lone pair electron in H_2O , or p orbitals from F^- ions directed toward the metal ion.

The d_{xy} , d_{yz} , and d_{xz} orbitals, whose lobes are oriented at 45 degrees to the bond axes, do not overlap with the ligand orbitals and are nonbonding.

Ligand-to-metal ($L \rightarrow M$) σ donation



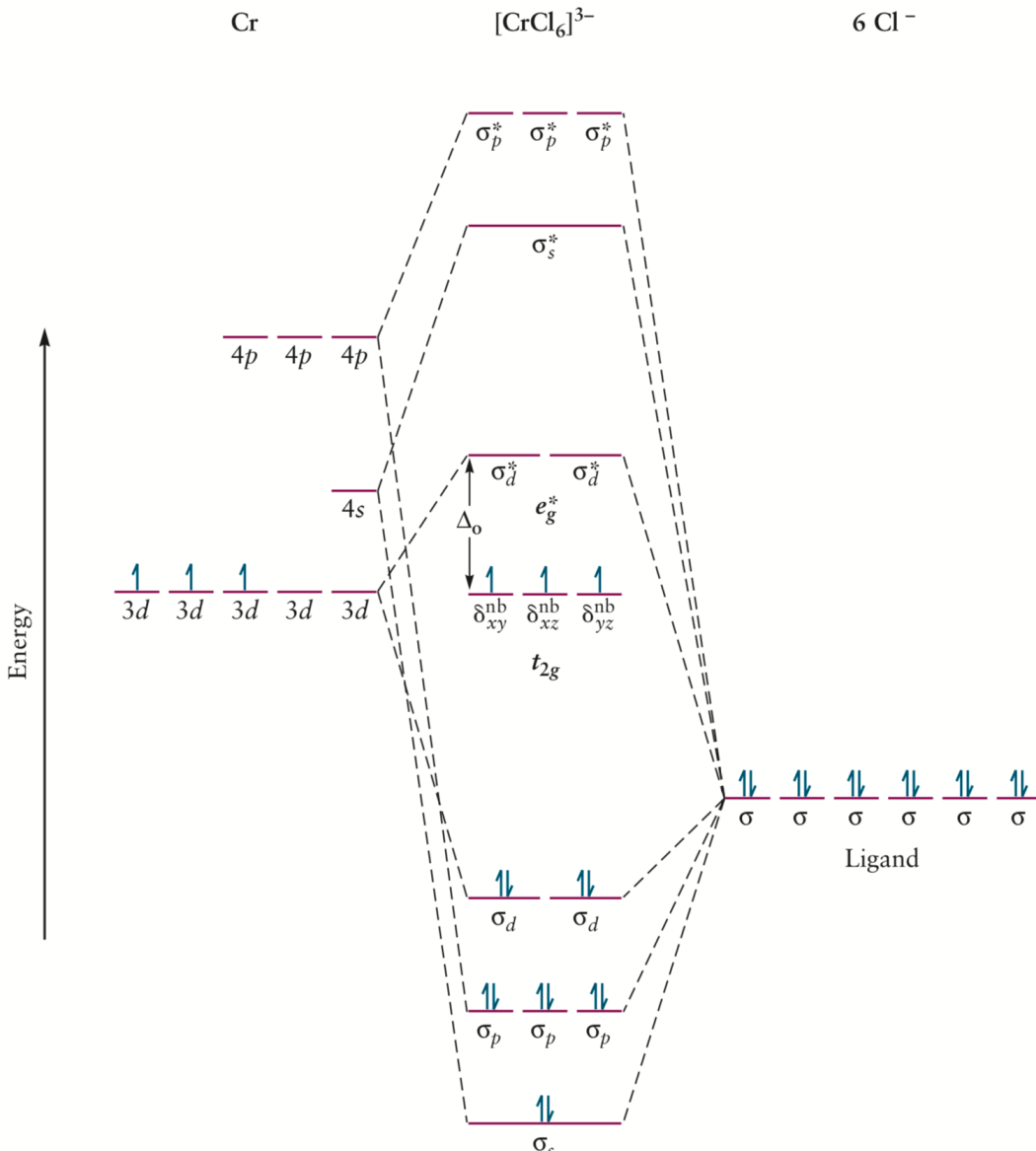
Bonding in Coordination Complexes: Molecular Orbital Theory



σ_d orbital derived from the metal d_{z^2} orbital delocalized along the z-axis and the metal $d_{x^2-y^2}$ orbital delocalized over the x-y plane.

σ_p orbitals are each delocalized along one of the Cartesian axes.

σ_s orbital is delocalized over the entire complex.



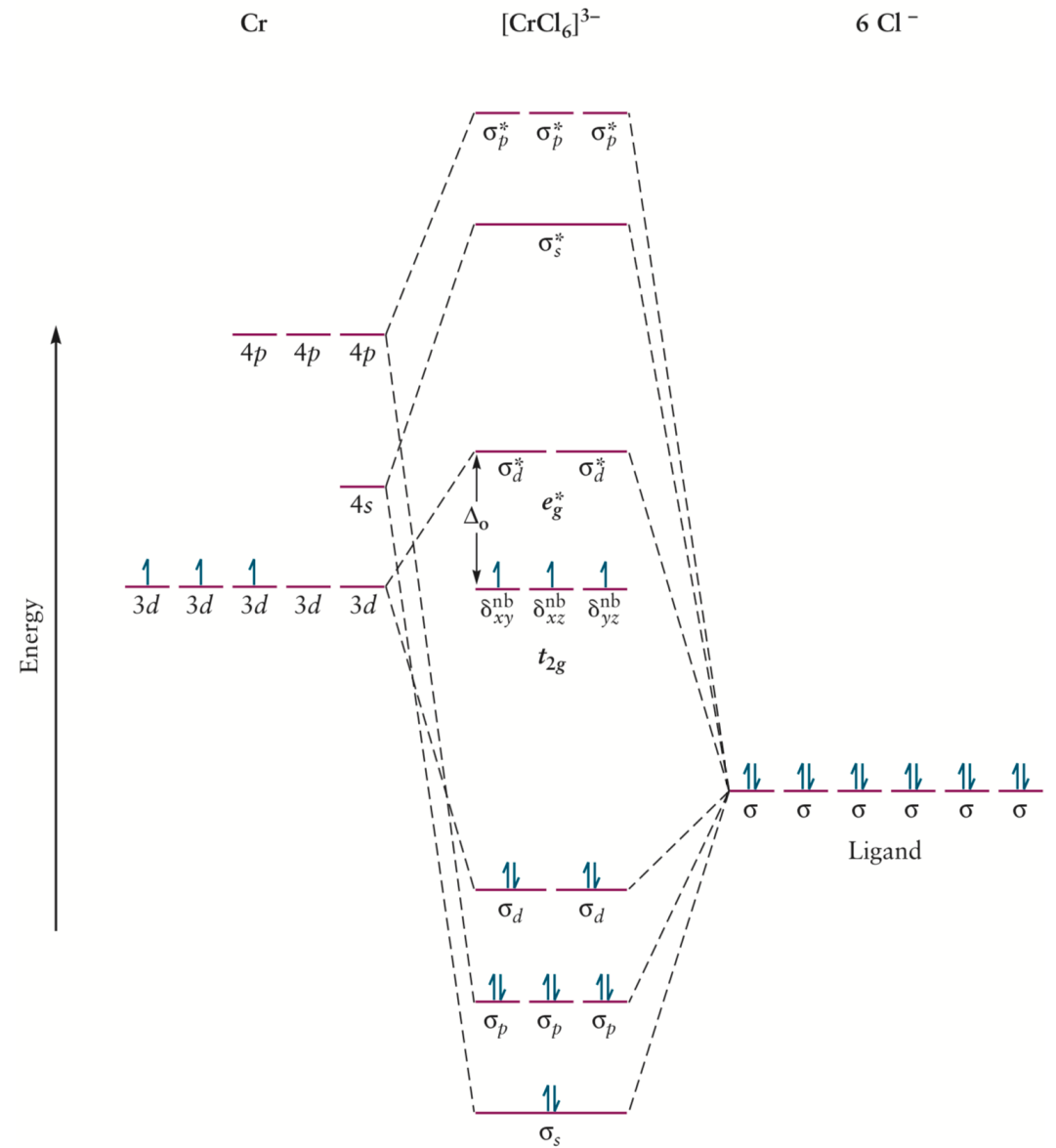
Ligand-to-metal ($L \rightarrow M$) σ donation

Bonding in Coordination Complexes: Molecular Orbital Theory

Ligand-to-metal ($L \rightarrow M$) σ donation

Strong interactions decrease the energy of **(stabilize)** the **bonding σ_d** orbital but increase the energy of **(de-stabilize)** the **σ_d^* orbital**.

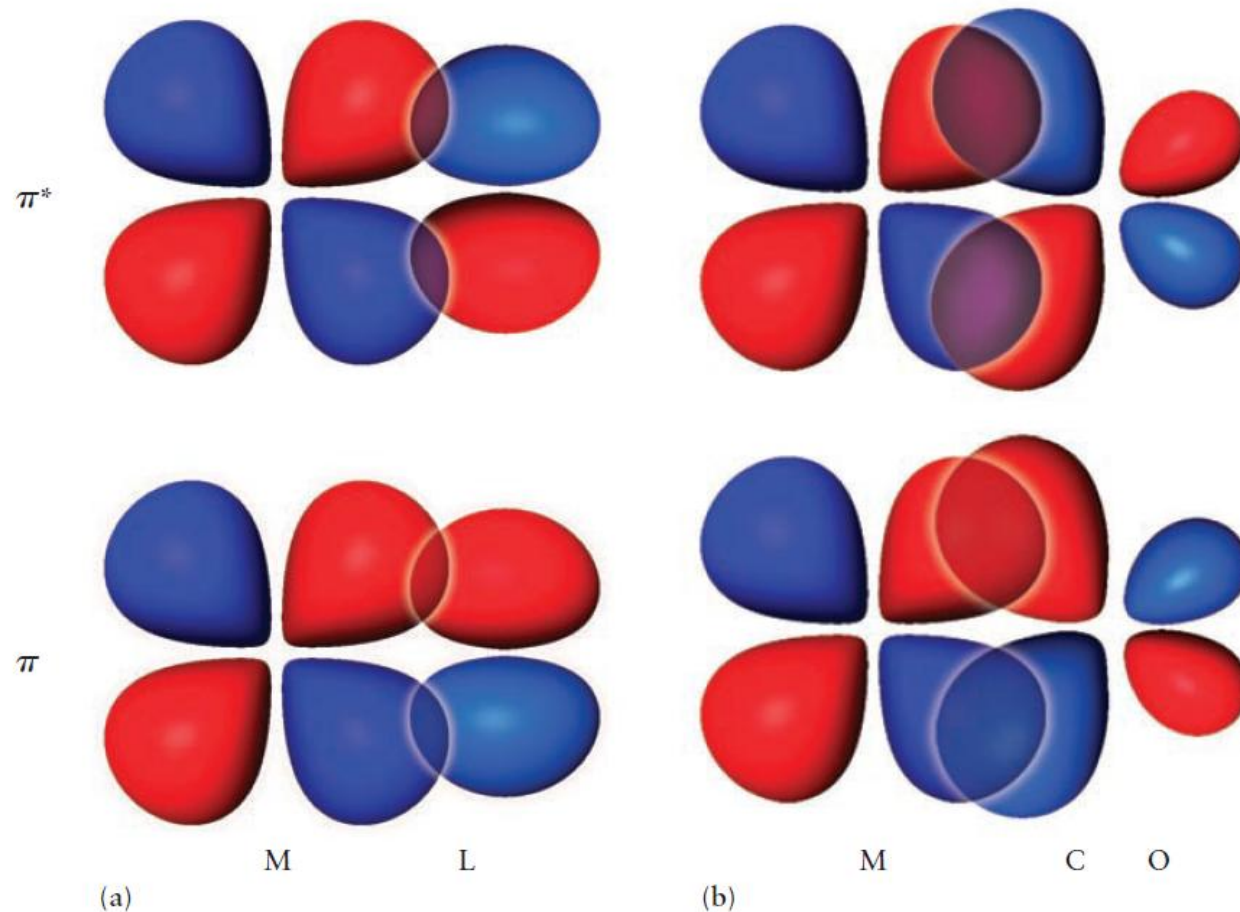
This description explains the increase in Δ_o as the strength of the metal–ligand interaction increases.



Bonding in Coordination Complexes: Molecular Orbital Theory

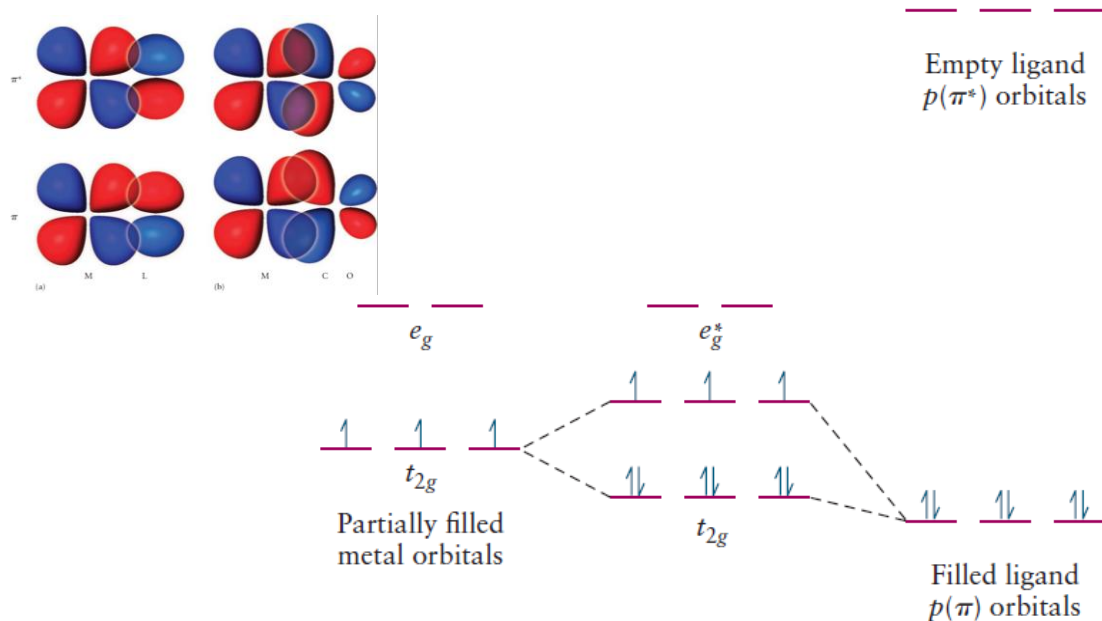
The different strengths of ligands such as the halides and those containing unfilled antibonding π orbitals like CO and CN^- **cannot be explained considering σ bonding alone.**

FIGURE 8.29 (a) Bonding and antibonding π orbitals formed by "side-by-side" overlap of a metal d orbital with an atomic ligand p orbital. (b) Bonding and antibonding π orbitals formed by "side-by-side" overlap of a metal d orbital with a molecular ligand π^* orbital.



$L \rightarrow M$ π donation:

- metals in high oxidation states (few d electrons, empty d orbitals)
- ligands with filled orbitals whose energies lie close to those of the empty d orbitals (the halides, water, OH and ammonia)

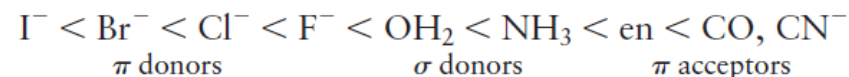


(a)

π donor ligands

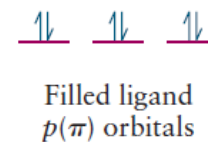
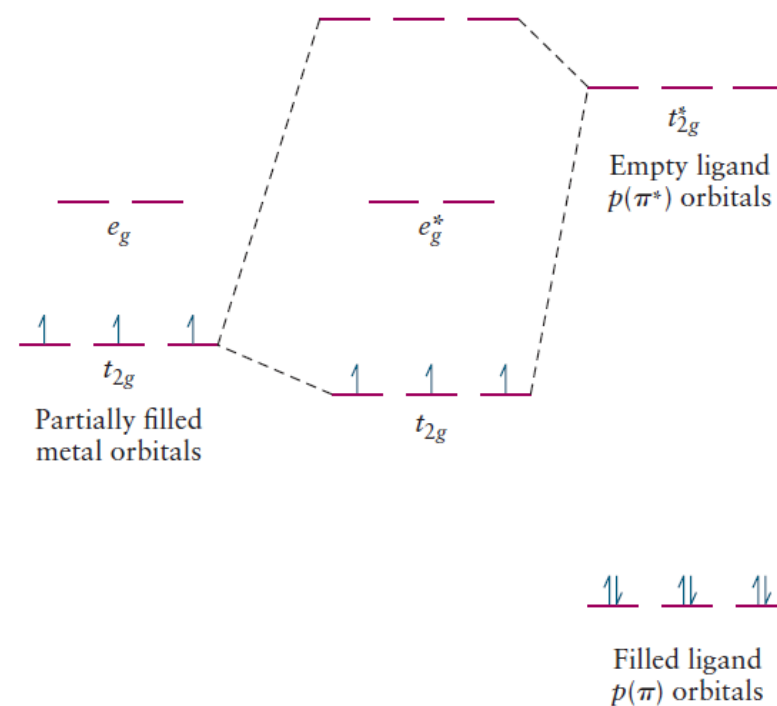
(b)

π acceptor ligands



$M \rightarrow L$ π backbonding

- metals in low oxidation states (many d electrons, few unfilled d orbitals)
- ligands with empty orbitals whose energies lie close to those of the filled d orbitals (d orbitals of P or S, or π^* orbitals in ligands such as CO and CN)

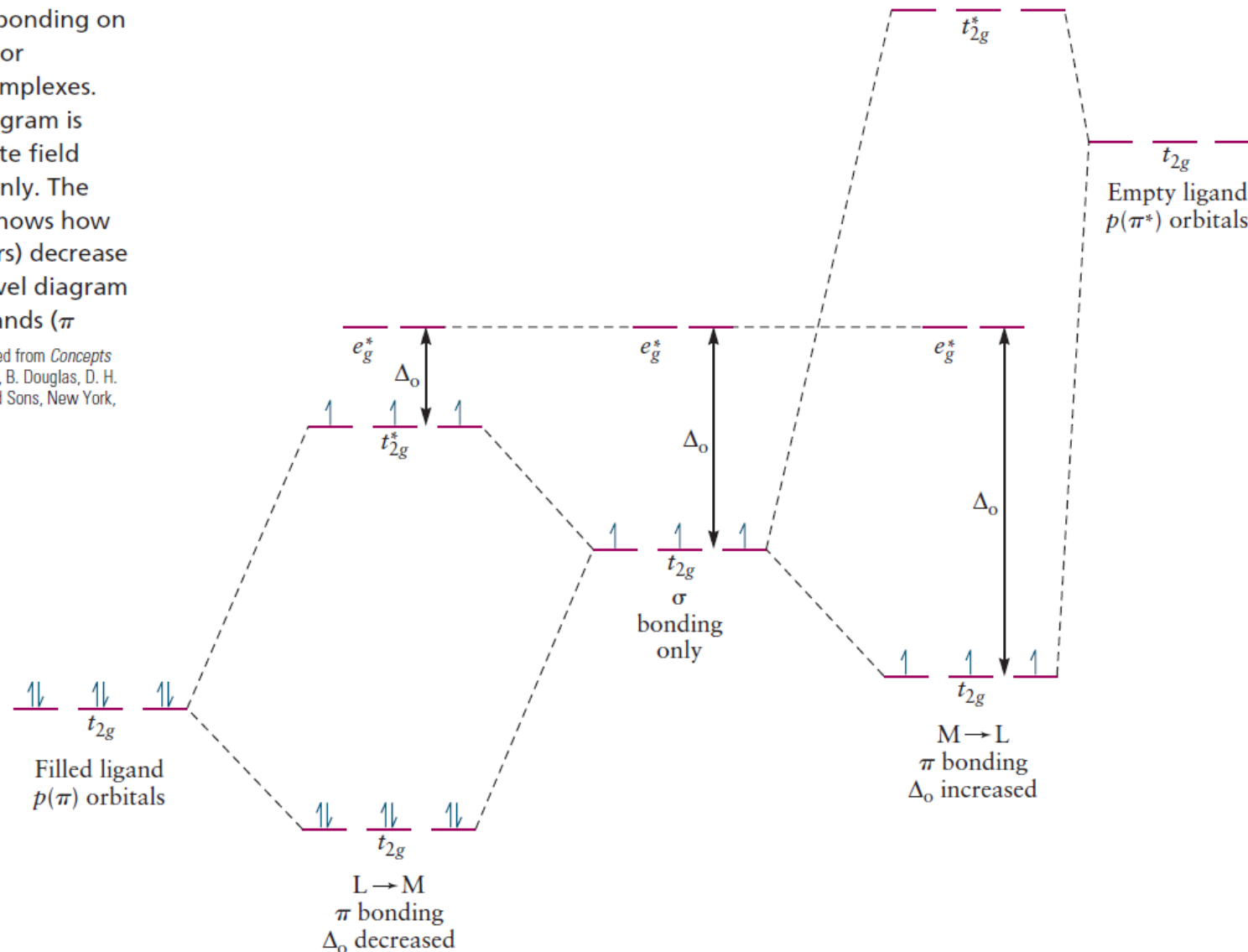


Bonding in Coordination Complexes: Molecular Orbital Theory

FIGURE 8.32 Effect of π bonding on the energy-level structure for octahedral coordination complexes.

The center energy-level diagram is appropriate for intermediate field ligands that are σ donors only. The left energy-level diagram shows how weak field ligands (π donors) decrease Δ_o , and the right energy-level diagram shows how strong field ligands (π acceptors) increase Δ_o .

(Adapted from *Concepts and Models of Inorganic Chemistry*, 2nd edition, B. Douglas, D. H. McDaniel, and J. J. Alexander, John Wiley and Sons, New York, 1983, p. 293.)

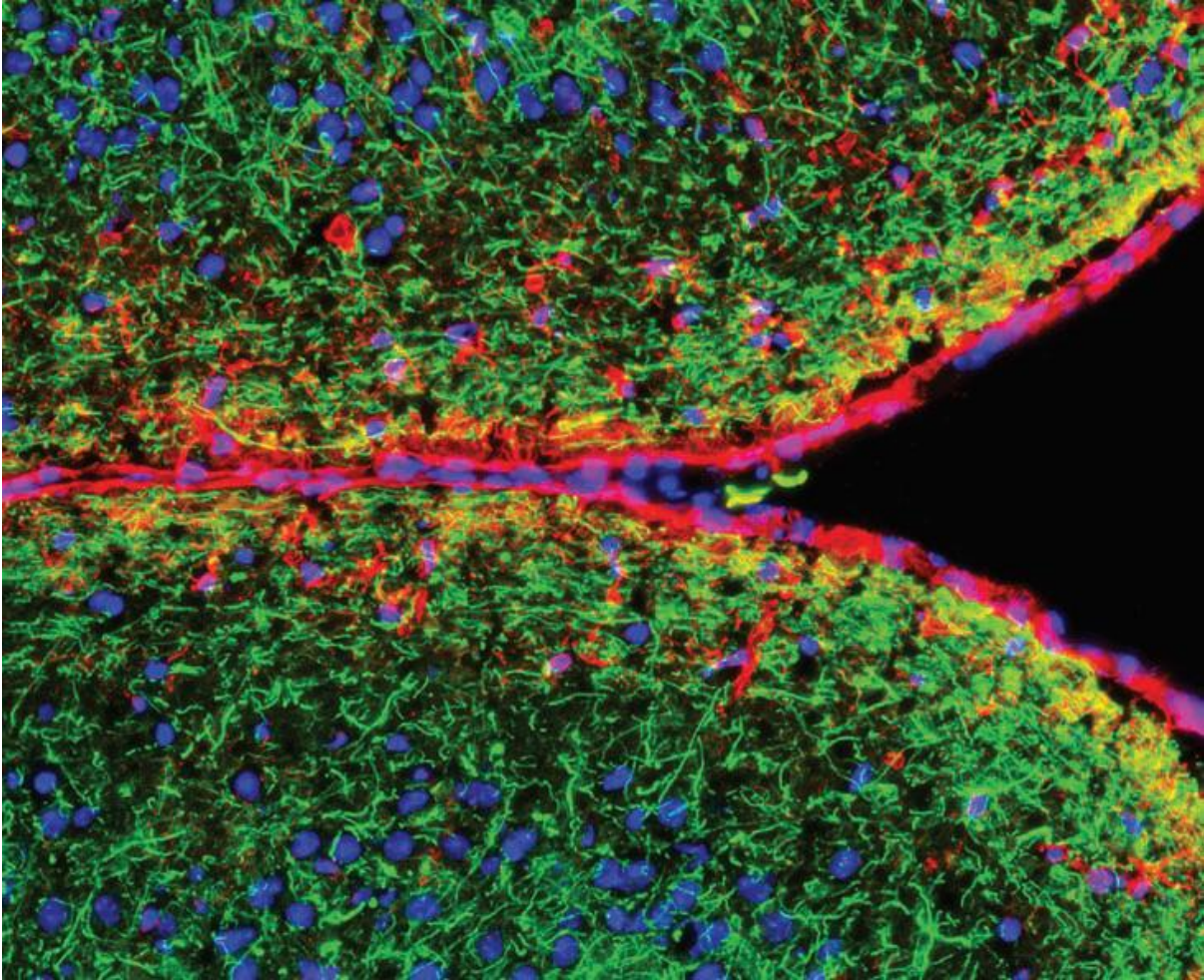


Major points of today's lecture

- **Molecular spectroscopy and photochemistry**

1. **Introduction to Molecular Spectroscopy**
2. **Experimental Methods in Molecular Spectroscopy**
3. **Rotational and Vibrational Spectroscopy**
4. **Nuclear Magnetic Resonance Spectroscopy**
5. **Electronic Spectroscopy and Excited State Relaxation Processes**
6. **Introduction to Atmospheric Chemistry**
7. **Photosynthesis**

Introduction to Molecular Spectroscopy



Spectroscopy (光谱): the interactions between electromagnetic radiation and matter.

Absorbed 吸收

Emitted 发射

Scattered 散射

Introduction to Molecular Spectroscopy

Spectroscopy: the intensity of radiation absorbed, emitted, or scattered vs. frequency

• **Atomic absorption and emission spectra**  $\Delta E = h\nu_i$

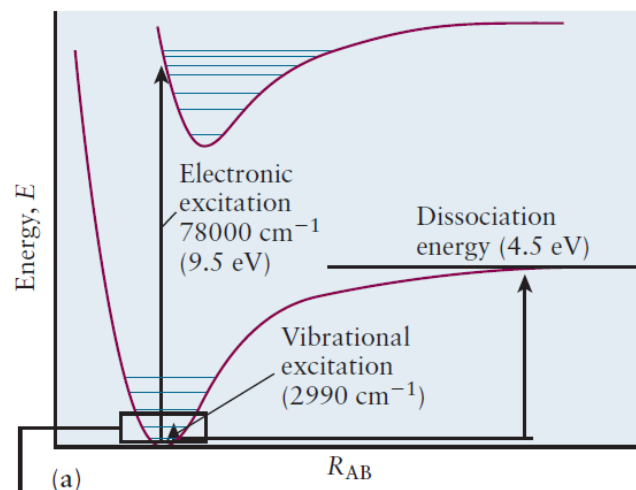
• **Raman scattering: transitions between molecular energy levels**  $\Delta E = h\nu_i - h\nu_s$

Inelastic (非弾性) light scattering: photons can either gain or lose energy in inelastic collisions with molecules

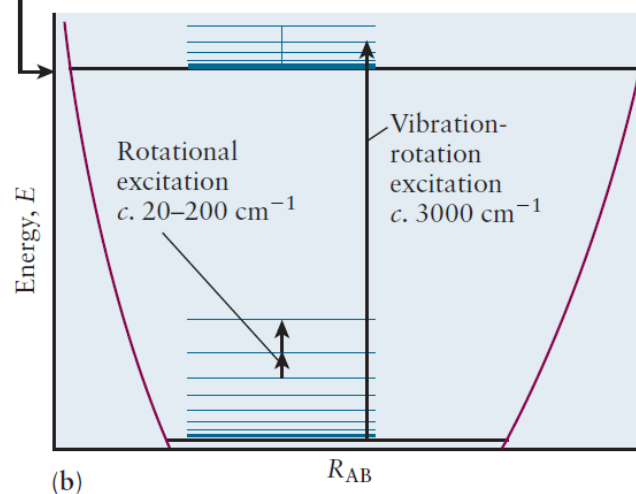
Introduction to Molecular Spectroscopy

Born–Oppenheimer approximation: slow nuclei, fast electrons.

$$E_{\text{total}} = E_{\text{trans}} + E_{\text{rot}} + E_{\text{vib}} + E_{\text{el}}$$



Absorption from the ground electronic state to an excited electronic state



Vibrational levels in the ground electronic state showing pure rotation, pure vibration, and vibration-rotation transitions

Introduction to Molecular Spectroscopy

- **Bond lengths and bond angles:** microwave, infrared, and Raman spectroscopy
- **Bond force constants and effective reduced masses:** vibrational spectroscopy
- **Identification of functional groups and their relative locations in molecules:** vibrational spectroscopy and nuclear magnetic resonance spectroscopy

T A B L E 20.1					
Spectroscopic Methods					
Spectral Region	Frequency	Wavelength	Transition Energies	Excitations	Information Obtained
Radio waves	$10^7\text{--}10^9$ Hz	30–0.3 m	$0.0033\text{--}0.033\text{ cm}^{-1}$	Nuclear spin	Bonding environment, molecular structure
Microwave, far infrared	$10^9\text{--}10^{12}$ Hz	0.3 m–300 μm	$0.033\text{--}33\text{ cm}^{-1}$	Rotational	Bond lengths and bond angles
Near infrared	$10^{12}\text{--}10^{14}$ Hz	300 μm –3 μm	$33\text{--}3300\text{ cm}^{-1}$	Vibrational	Bond force constants and effective reduced mass
Visible, ultraviolet	$10^{14}\text{--}10^{17}$ Hz	3000 nm–3 nm	$3300\text{--}330,000\text{ cm}^{-1}$	Valence electrons	Electronic energy levels and molecular environment
X-ray	$10^{17}\text{--}10^{19}$ Hz	3 nm–0.3 Å	413–41,300 eV	Core electrons	Ionization energies and atomic environment

Introduction to Molecular Spectroscopy

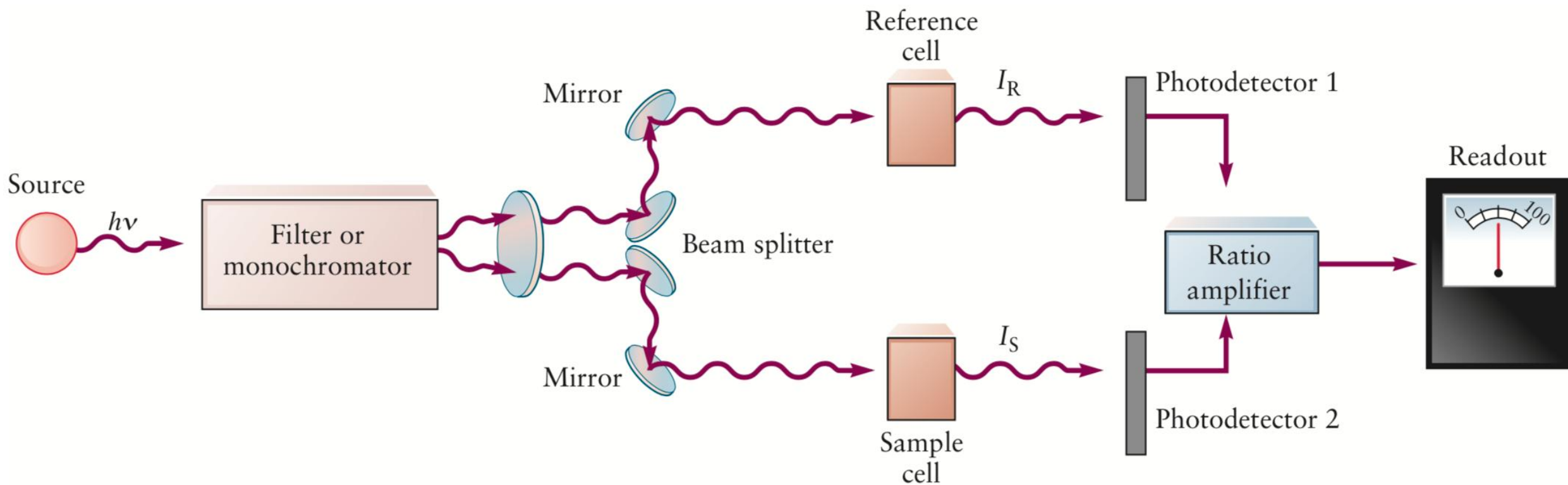


FIGURE 20.2 Schematic of double-beam spectrophotometer. Incoming light passes through a reference cell identical to the sample cell.

Intensities of Spectral Transitions

The intensity of the transmitted beam: **Beer–Lambert law**

$$I_t = I_0 e^{-\alpha \ell}$$

I_0 : intensity of the incident beam

I_t : intensity of the transmitted beam

ℓ : path length

α : absorption coefficient for the sample

The Beer–Lambert law (first order kinetics): a constant fraction of the light is absorbed per unit path length and that fraction is independent of the intensity of the incident radiation

$$I_t = I_0 \exp[-(N/V)\sigma \ell]$$

N/V : the concentration of light-absorbing molecules

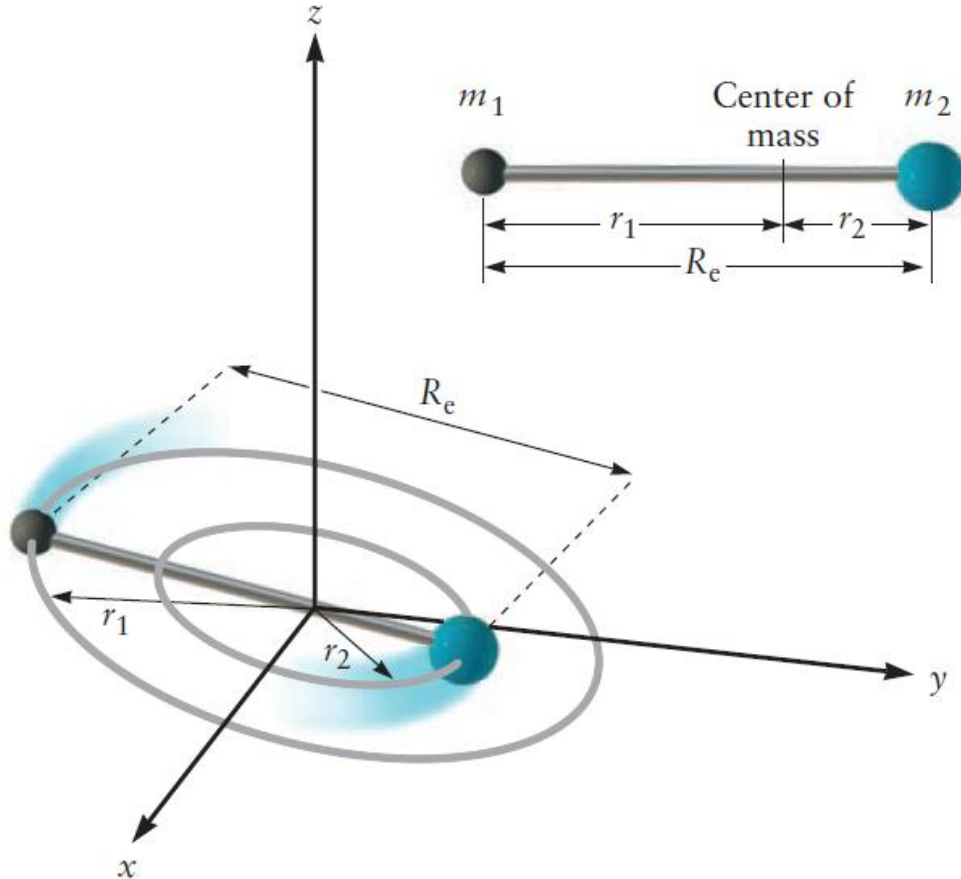
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7. Photosynthesis

Diatomic Molecules: Rotational Spectroscopy

Diatomic molecules: **linear rigid rotor model 刚性转子** (the bond length is fixed; the molecule rotates about an axis that is perpendicular to the internuclear axis and passes through its center of mass)



The rotational motion is described by a **single moment of inertia 单惯性矩**

$$I = \mu R_e^2$$

R_e : equilibrium bond length

$\mu = m_1 m_2 / (m_1 + m_2)$, reduced mass

$$r_1 = \left(\frac{m_2}{m_1 + m_2} \right) R_e \text{ and } r_2 = \left(\frac{m_1}{m_1 + m_2} \right) R_e$$

Diatomic Molecules: Rotational Spectroscopy

The **wave functions for the rigid rotor** are identical to the angular parts of the hydrogen atom wave functions (the angular momentum is quantized).

The square of the angular momentum
angular momentum projection along the z-axis

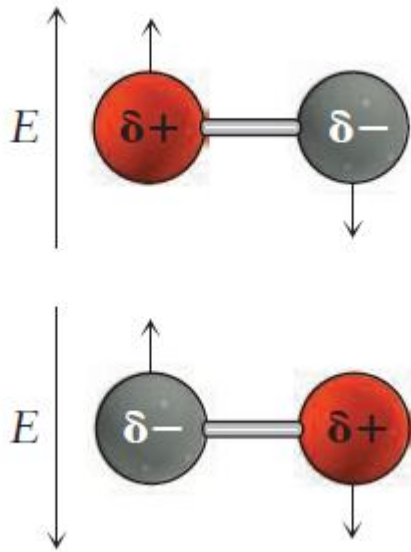
$$J^2 = J(J + 1)(h/2\pi)^2 \quad J = 0, 1, 2, 3...$$
$$J_z = M_J(h/2\pi) \quad M_J = -J, -J + 1, \dots, 0, \dots, J - 1, J$$

The energy levels for the linear rigid rotor

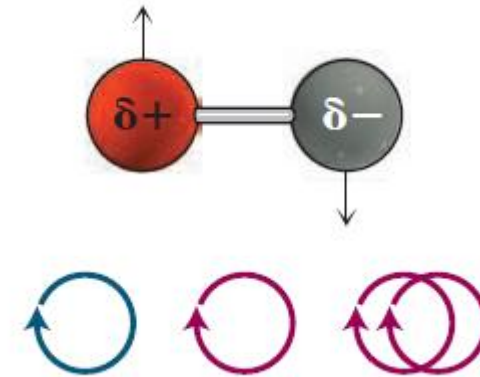
$$E_J = hc\tilde{B}J(J + 1) \quad \tilde{B} = (h/8\pi^2 cI)$$
$$E_J = hBJ(J + 1) \quad B = (h/8\pi^2 I)$$

Diatomic Molecules: Rotational Spectroscopy

Conservation of angular momentum: $\Delta J = \pm 1$ for microwave absorption and emission.



oscillating dipole moment of a heteronuclear diatomic molecule as it rotates



The angular momentum of a molecule decreases by **one unit** when a photon is emitted, the angular momentum being **carried by the photon**.

Diatomic Molecules: Rotational Spectroscopy

Conservation of energy for microwave absorption and emission.

selection rule: the frequencies of the allowed transitions in wave numbers

$$\begin{aligned}\tilde{\nu} &= \tilde{B}[J_f(J_f + 1) - J_i(J_i + 1)] \\ &= \tilde{B}[(J_i + 1)(J_i + 2) - J_i(J_i + 1)] \\ &= 2\tilde{B}(J + 1)\end{aligned}$$

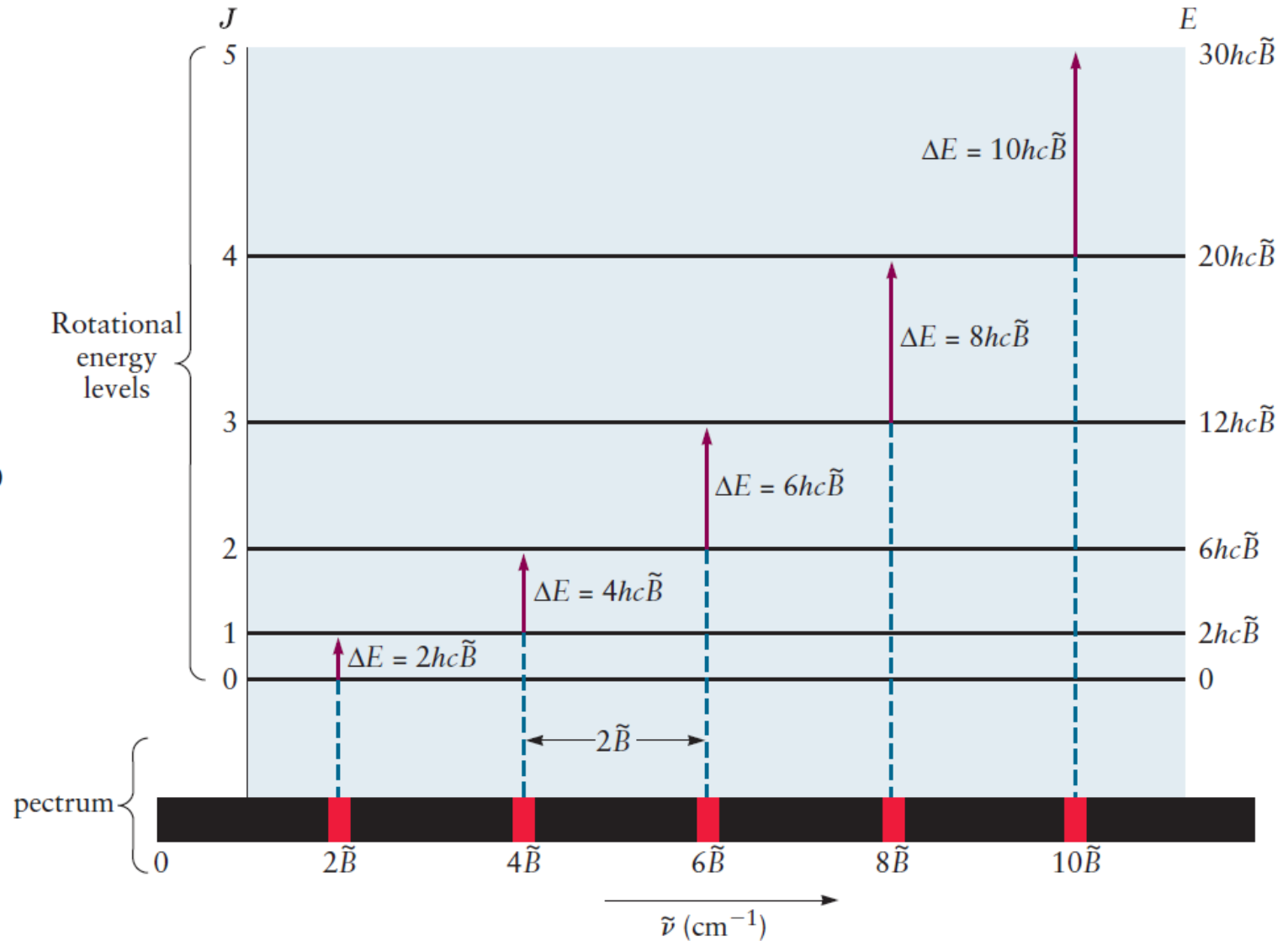
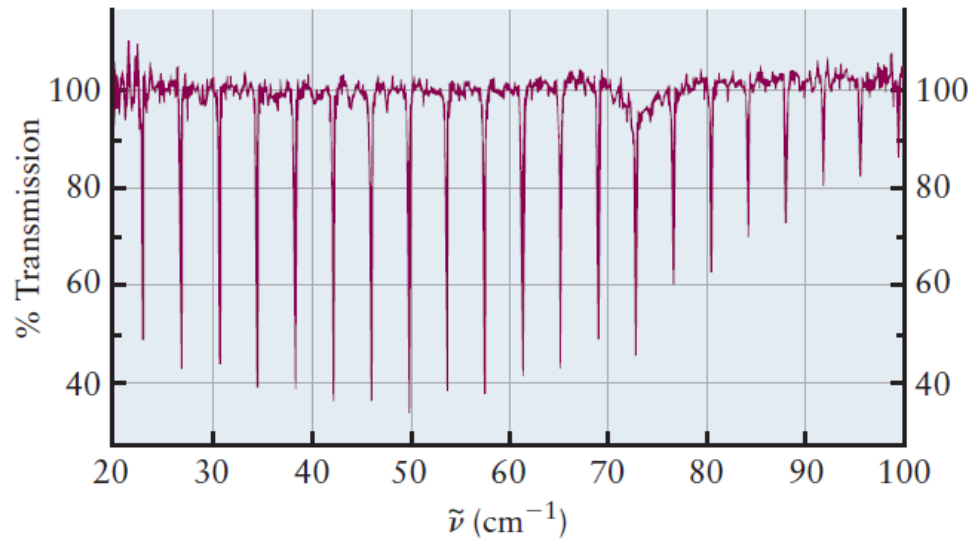
J_i and J_f represent the rotational quantum numbers of the initial and final states

The microwave absorption spectrum of gaseous NaH (isotope: $^{23}\text{Na}^1\text{H}$) has been measured experimentally; microwaves with wavelengths of 1.02 mm excite the transition from $J = 0$ to $J = 1$. Calculate the bond length of the NaH molecule. Use isotopic masses provided in Table 19.1.

Diatomic Molecules: Rotational Spectroscopy

The allowed transition frequencies

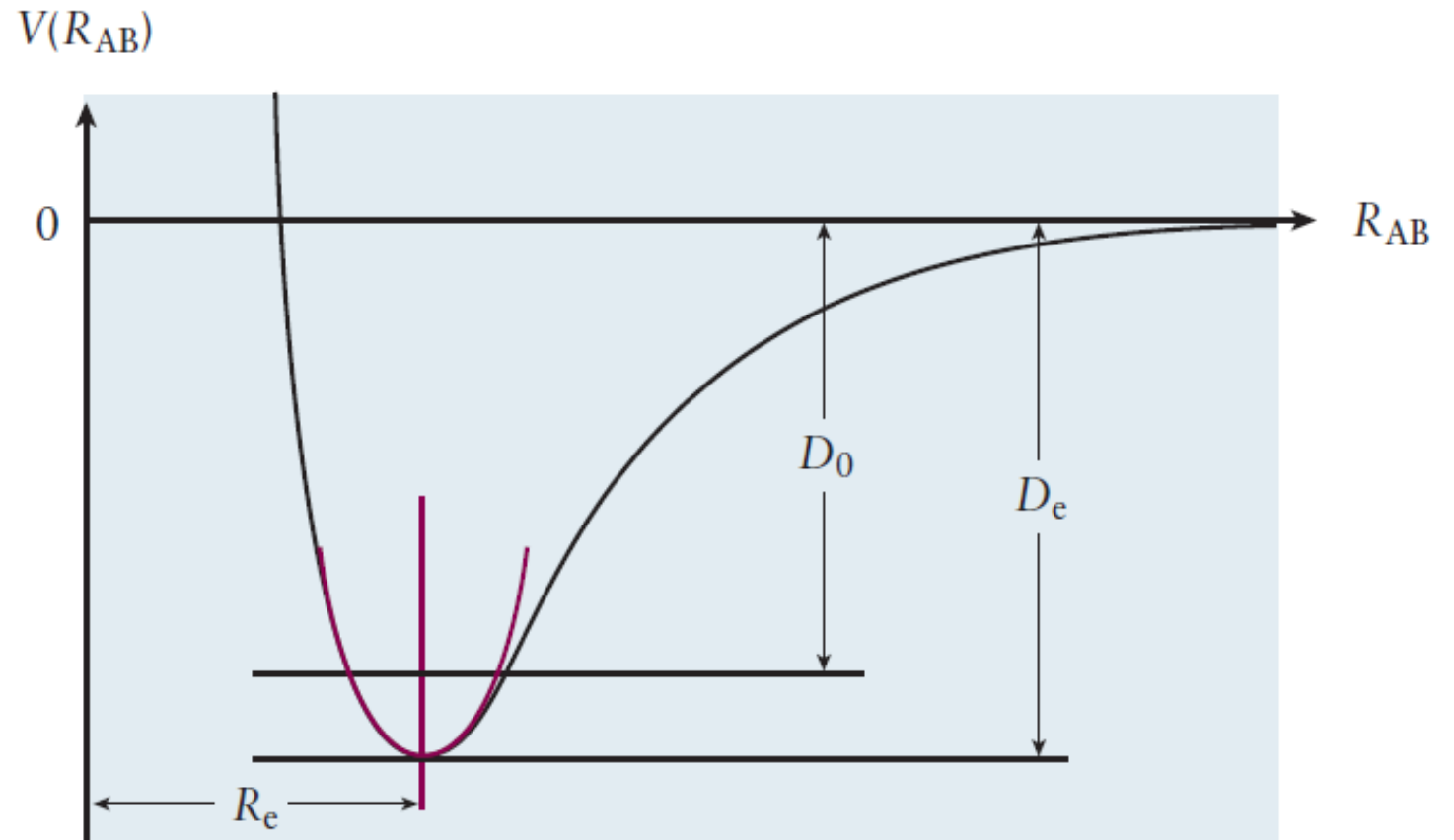
$$\begin{aligned}\Delta\tilde{\nu} &= \tilde{B}[J_f(J_f + 1) - J_i(J_i + 1)] \\ &= \tilde{B}[(J_i + 1)(J_i + 2) - J_i(J_i + 1)] \\ &= 2\tilde{B}(J + 1)\end{aligned}$$



Diatomic Molecules: Vibrational Spectroscopy

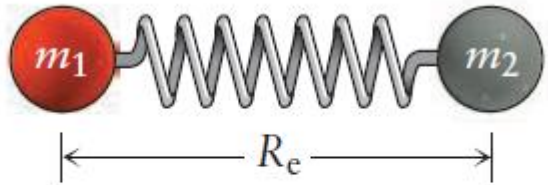
Vibrational Spectroscopy: estimate the “stiffness” of the bond (a measure of the bond strength)

FIGURE 20.14 Potential energy curves for a diatomic molecule (black) and a harmonic oscillator (red) that have the same equilibrium bond length R_e , bond dissociation energy D_e , and similar curvature near the minimum.



Diatomic Molecules: Vibrational Spectroscopy

two masses connected by a spring



Diatomic molecule represented by a ball and spring model.

reduced mass of the system $\mu = m_1 m_2 / (m_1 + m_2)$

Hooke's law $F(R - R_e) = -k(R - R_e)$

k: force constant (measure the stiffness of the bond)

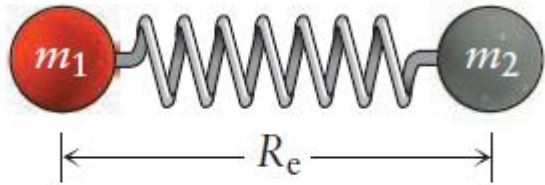
$$V(R - R_e) = \frac{1}{2} k(R - R_e)^2$$

Harmonic oscillator approximation: a characteristic frequency

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

Diatomic Molecules: Vibrational Spectroscopy

two masses connected by a spring



Diatomic molecule represented by a ball and spring model.

one-dimensional Schrödinger equation with the harmonic oscillator potential energy term added to the kinetic energy term.

$$-\frac{h^2}{8\pi^2 m} \frac{d^2 \psi(x)}{dx^2} + \frac{1}{2} kx^2 \psi(x) = E \psi(x)$$

The quantized energy levels of the harmonic oscillator

$$E_v = \left(v + \frac{1}{2}\right) h\nu$$

ν is its fundamental vibrational frequency and v is a quantum number **0, 1, 2, ...**

Diatomic Molecules: Vibrational Spectroscopy

E does not equal to 0 even at 0 K;
this motion is called **zero-point motion**;
the energy is called the **zero-point energy**.

TABLE 20.2

The First Four Harmonic Oscillator Wave Functions

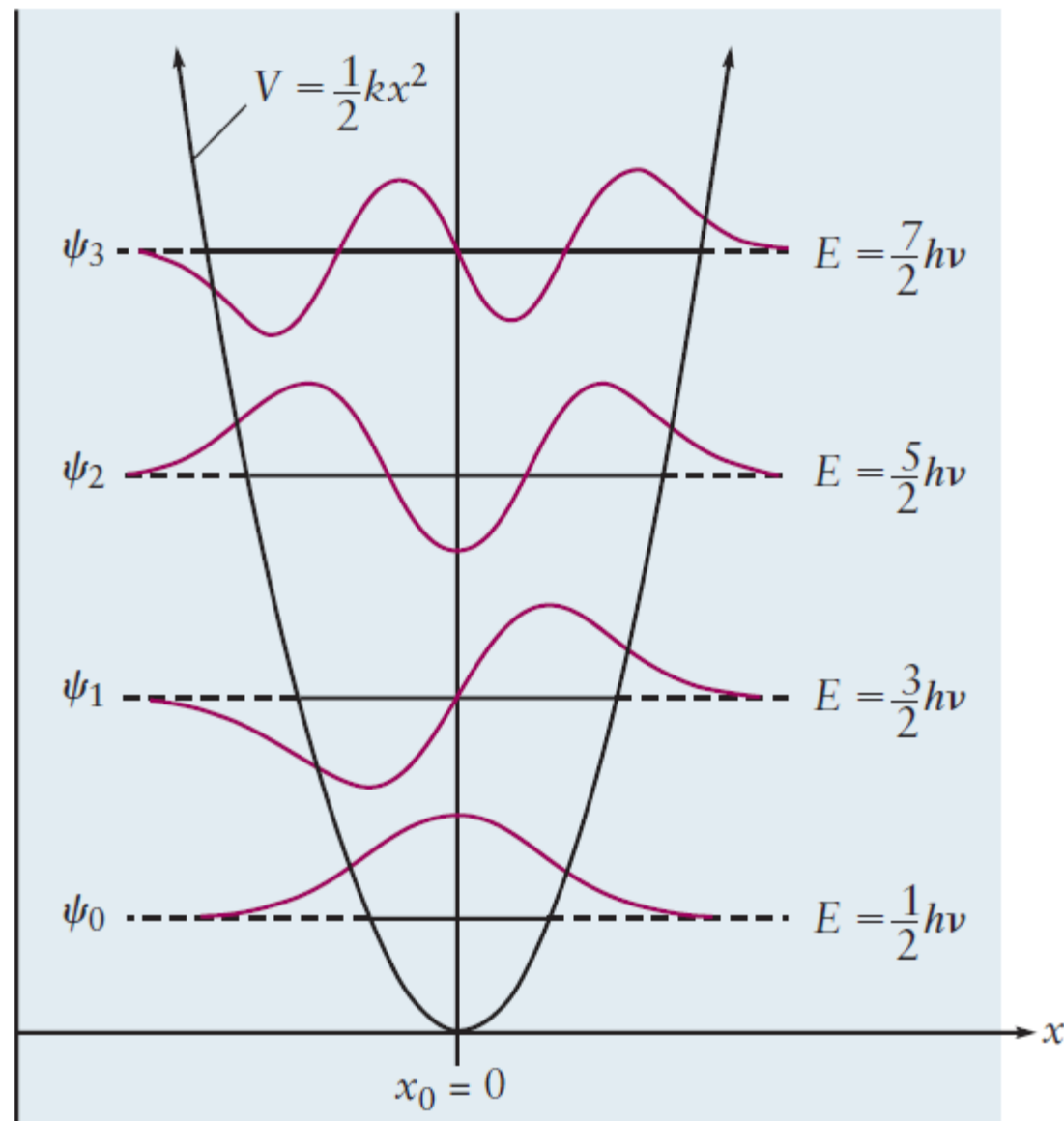
$$\alpha = \frac{2\pi}{h} (k\mu)^{1/2}$$

$$\psi_0(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$$

$$\psi_1(x) = \left(\frac{4\alpha^3}{\pi}\right)^{1/4} x e^{-\alpha x^2/2}$$

$$\psi_2(x) = \left(\frac{\alpha}{4\pi}\right)^{1/4} (2\alpha x^2 - 1) e^{-\alpha x^2/2}$$

$$\psi_3(x) = \left(\frac{\alpha^3}{9\pi}\right)^{1/4} (2\alpha x^3 - 3x) e^{-\alpha x^2/2}$$



Diatomic Molecules: Vibrational Spectroscopy

Energy differences between ground ($v = 0$) and first excited ($v = 1$) vibrational states: 2 kJ mol^{-1} to 40 kJ mol^{-1} ($>$ rotational energy differences; $<$ electronic energy differences)

1. These energy differences correspond to the absorption in the **infrared region** of the electromagnetic spectrum.
2. Strong absorption is observed only for transitions between states that differ by 1 in vibrational quantum number ($\Delta v = 1$).
3. Separation between adjacent energy levels is independent of quantum number: a single vibrational frequency to determine the force constant of the bond.

The infrared spectrum of gaseous NaH (isotope: $^{23}\text{Na}^1\text{H}$) was determined via absorption spectroscopy. A single absorption band was observed at $\lambda = 8.53 \mu\text{m}$. Calculate the vibrational force constant of the NaH molecule using the harmonic oscillator model using the masses of the isotopes provided in Table 19.1.

Major points of today's lecture

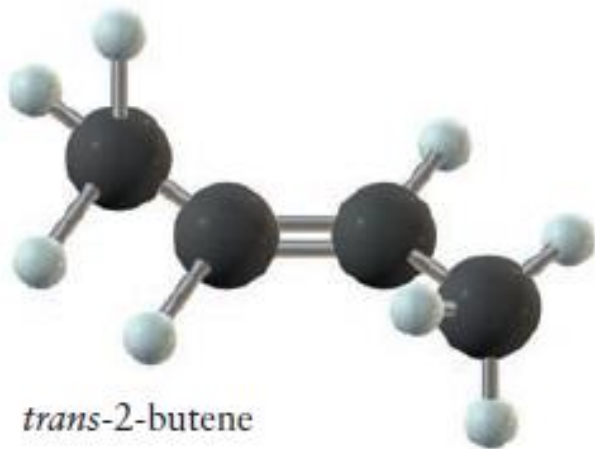
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The Alkenes and Alkynes



cis-2-butene

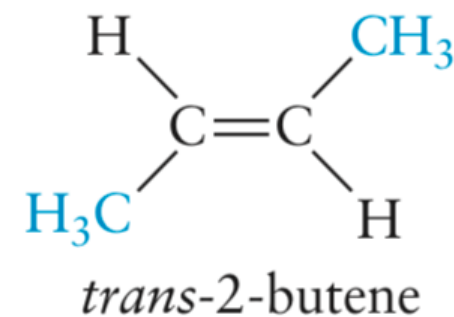
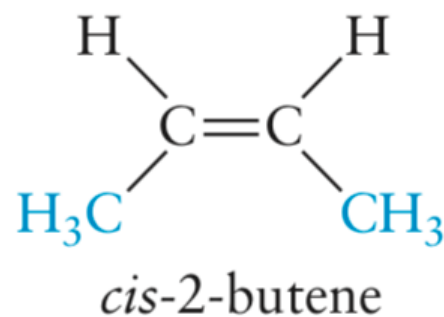
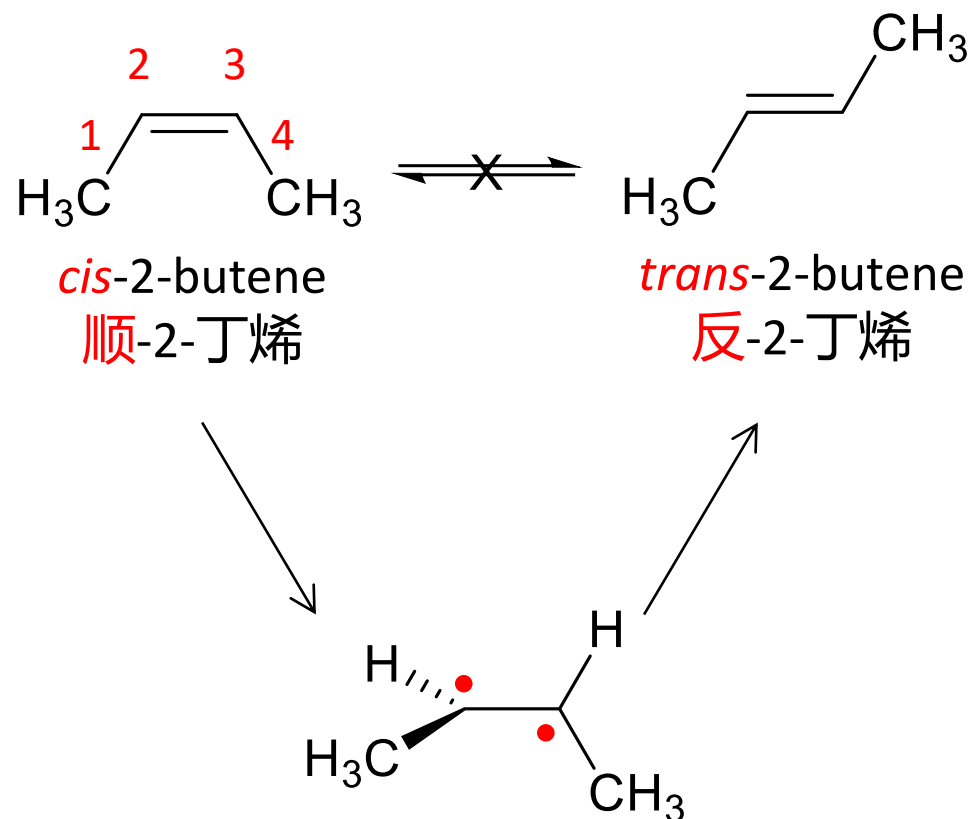


trans-2-butene

FIGURE 7.14 The two *cis-trans* isomers of 2-butene.

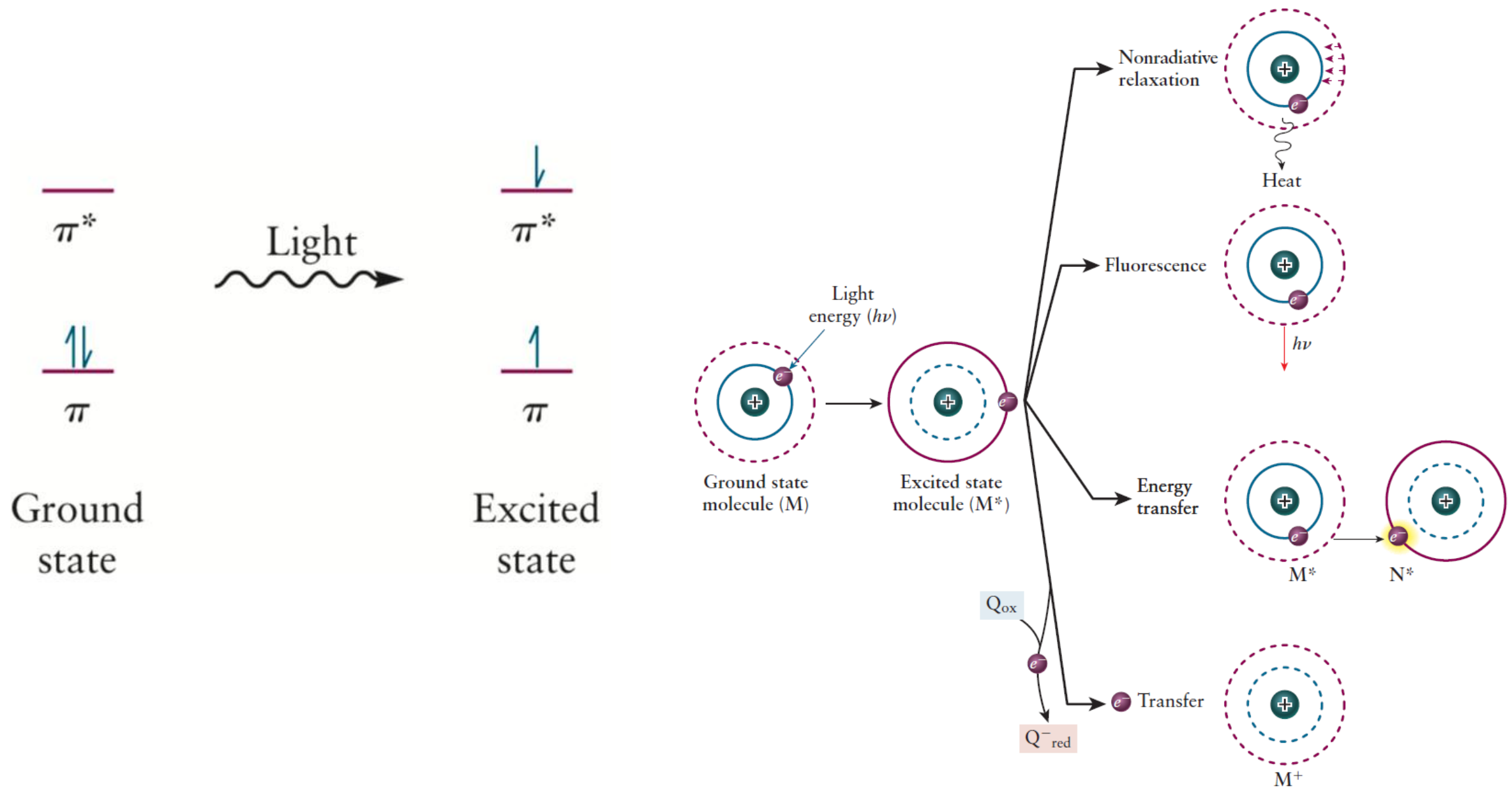
- Breaking a π bond costs a significant amount of energy: **both *cis* (顺式) and *trans* (反式) forms are stable**, and interconversion between the two is slow.
- The change from *trans* to *cis* can be accomplished by photochemistry without complete breaking of bonds.
- Molecules such as *trans*-2-butene can absorb ultraviolet light, which excites an electron from a π to a π^* MO.

The Alkenes and Alkynes

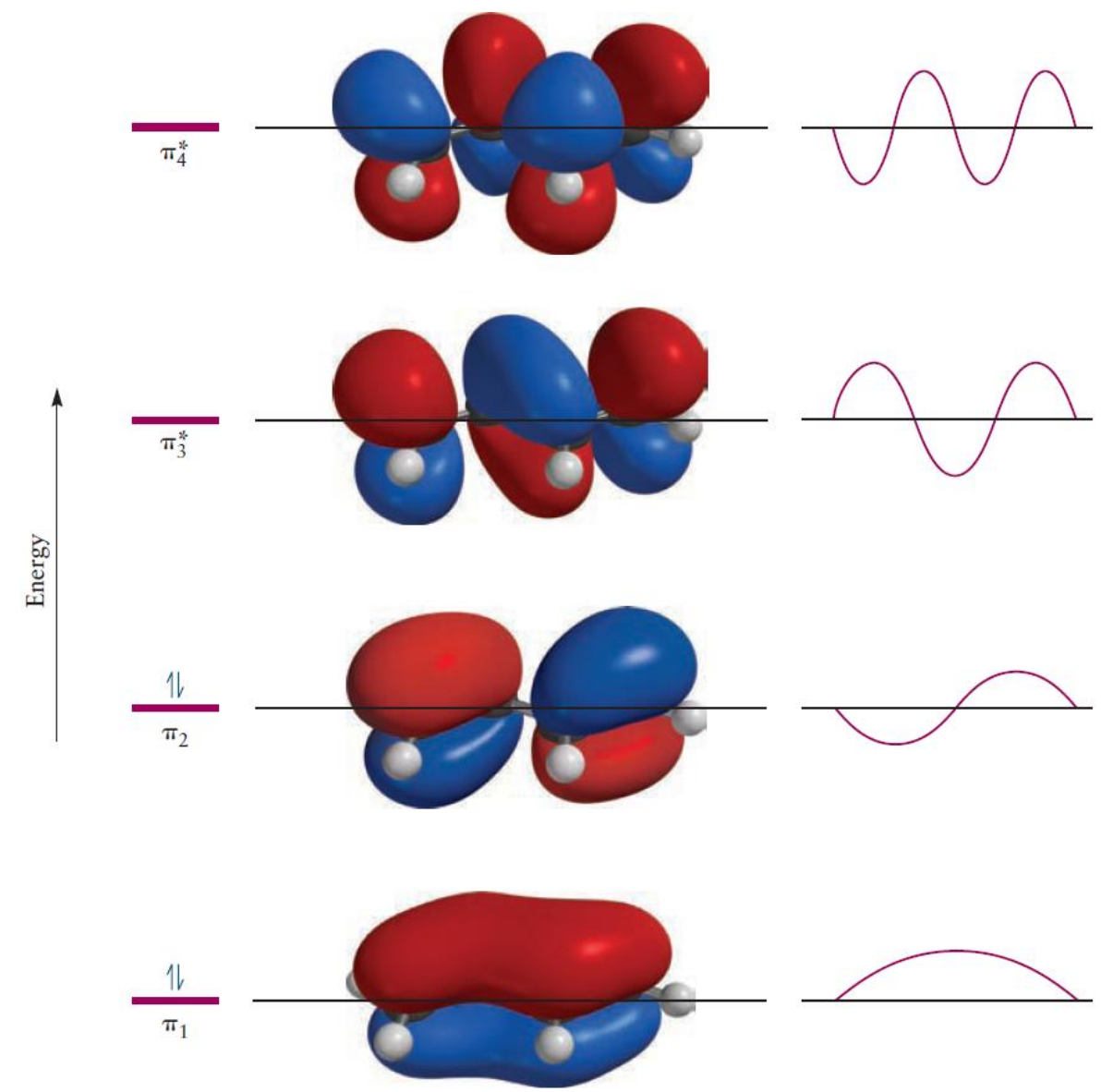
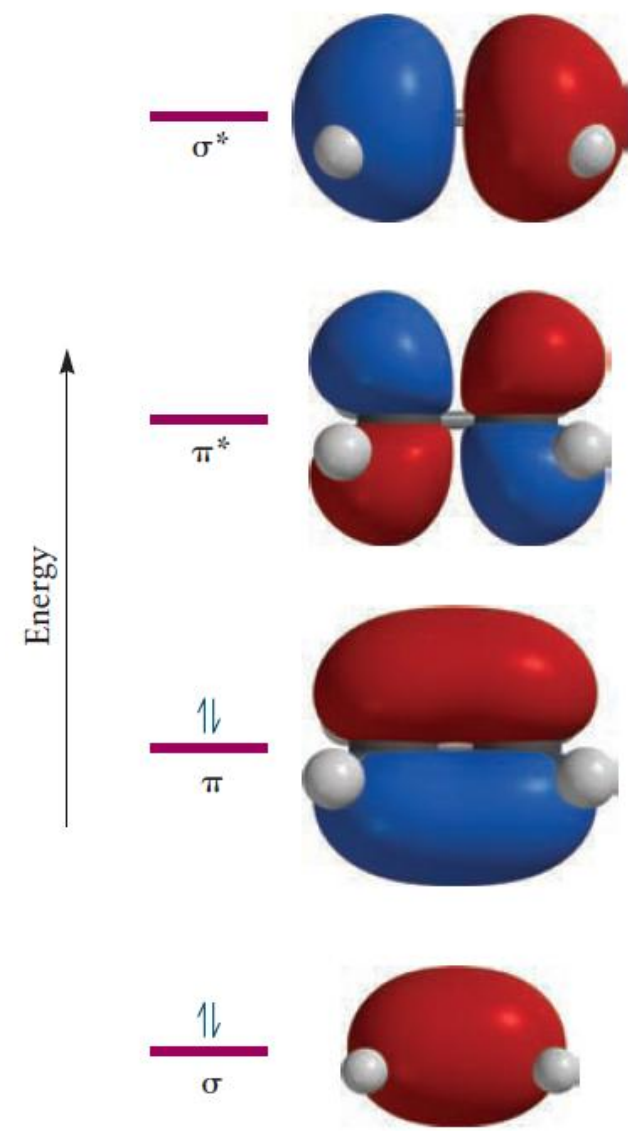


These compounds differ in **melting and boiling points, density**, and other **physical and chemical properties**

Excited State Relaxation Processes



Electronic Spectroscopy

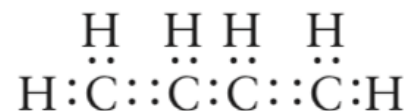


The Alkenes and Alkynes

Compounds with two double bonds are called dienes (二烯类).



1,3-pentadiene (1,3-戊二烯)



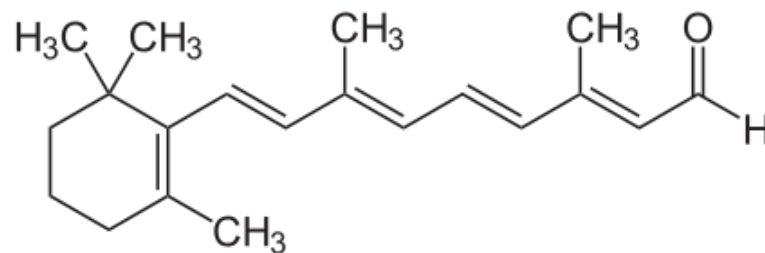
1,3-butadiene (1,3-丁二烯)

>2 double or triple bonds occur **close to each other** in a molecule: **a delocalized MO**.

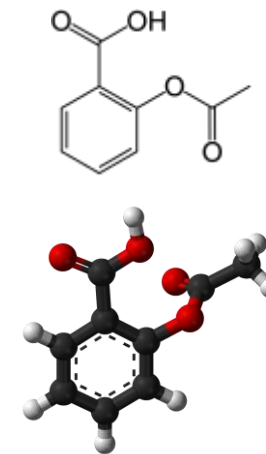
- All 4 carbon atoms have $\text{SN}=3$ (sp^2).
- The p_z orbitals have maximum overlap when the 4 carbon atoms lie in the same plane (predicted to be planar)

The Alkenes and Alkynes

Conjugate system (共轭体系)

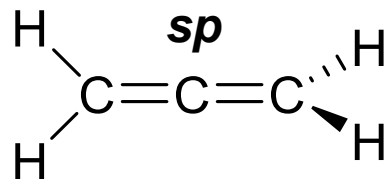
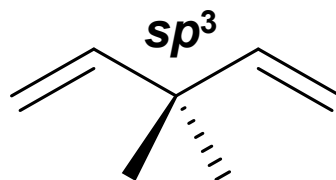


Retinal
视黄醛

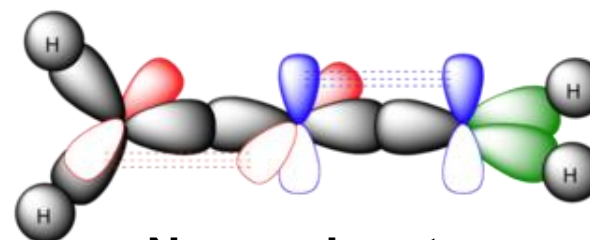
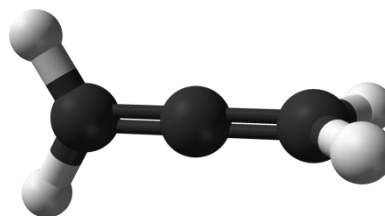


阿司匹林

Nonconjugate



Allene 丙二烯



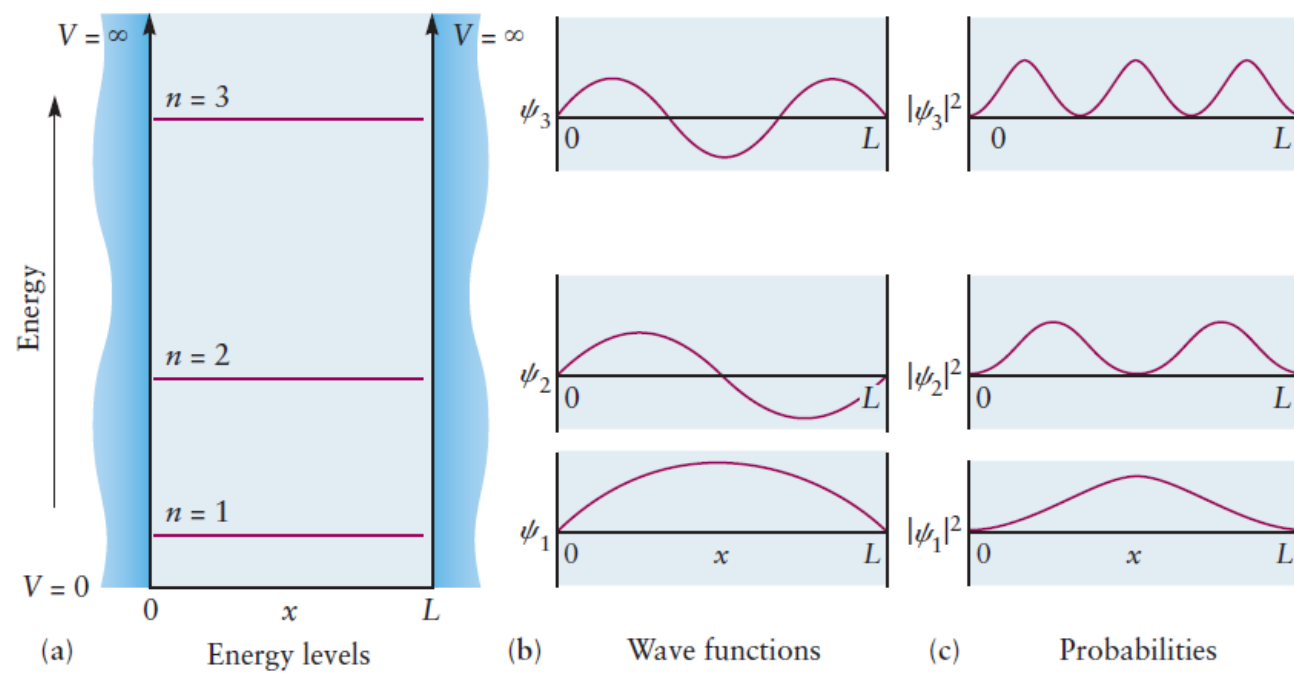
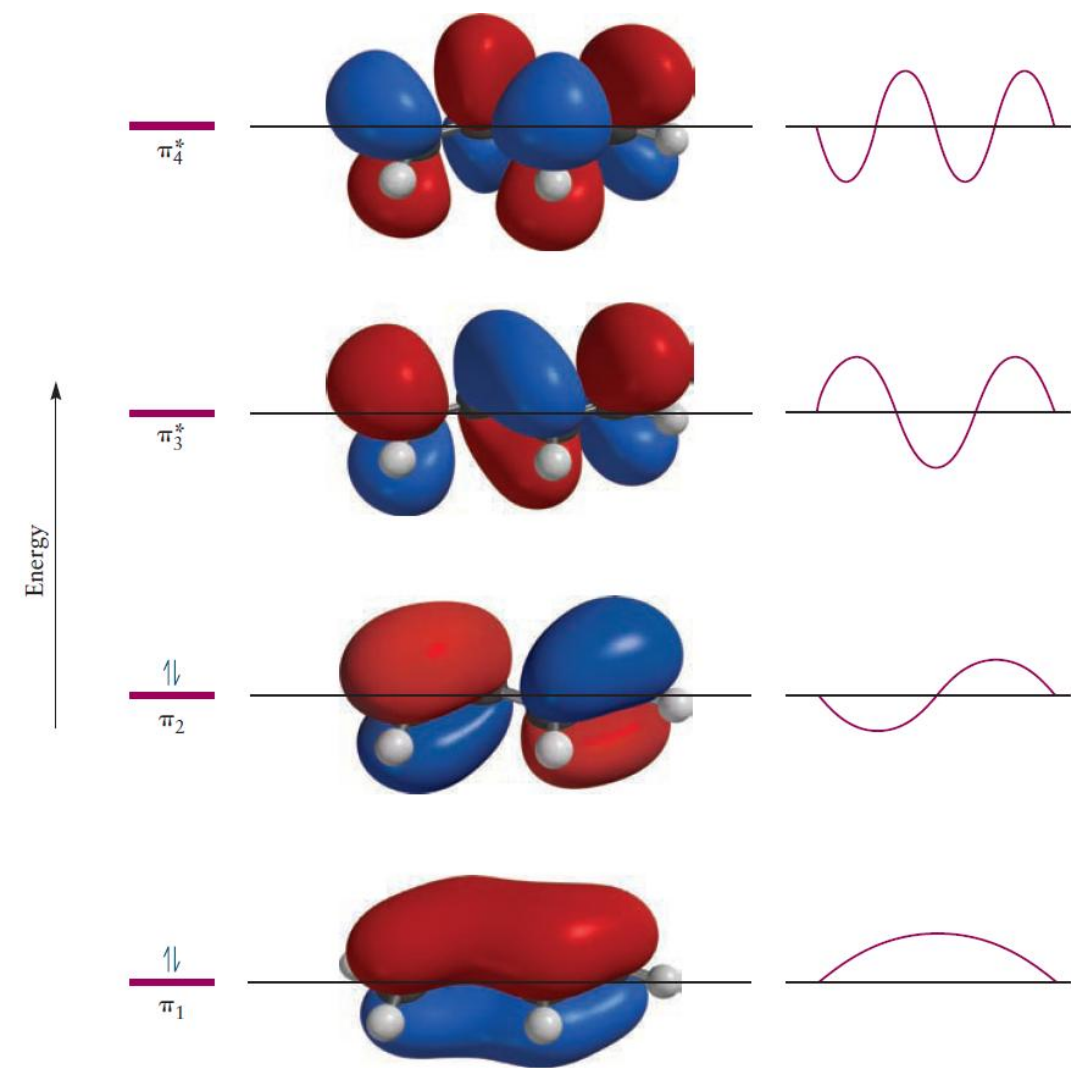
Nonconjugate

Question

EXAMPLE 4.7

The Lewis dot structure of 1,3-butadiene, $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ has two double bonds. Four electrons make two of the single bonds into double bonds, the extra pair of bonding electrons on the first, and the extra pair on the third of the carbon-carbon bonds. In a simple particle-in-a-box model for these four electrons, they may move freely along all three carbon-carbon bonds in the molecule but not past the end C atoms. If you draw a line along the carbon-carbon-carbon backbone of this molecule and label the center of one end C atom as $x = 0$, this corresponds to a one-dimensional box of length equal to the sum of the carbon-carbon bond lengths: C=C bond + C—C bond + C=C bond = $1.34 \text{ \AA} + 1.54 \text{ \AA} + 1.34 \text{ \AA} = 4.22 \text{ \AA}$. When two electrons are in the $n = 1$ state and two electrons are in the $n = 2$ state:

- (a) Calculate the most likely and the least likely positions of one of the $n = 2$ electrons.
- (b) Calculate the probability of finding this electron between the first two carbon atoms, between the second two carbon atoms, and between the third two carbon atoms.



$$E_n = \frac{n^2 h^2}{8mL^2} \quad n = 1, 2, 3, \dots$$

Solution

(a) Using Equation 4.36 the wave function for an electron in this $n = 2$ state is

$$\psi_2 = \sqrt{\frac{2}{4.22 \text{ Å}}} \sin\left(\frac{2\pi x}{4.22 \text{ Å}}\right) \quad \text{between } x = 0 \text{ Å and } x = 4.22 \text{ Å}$$

To find the probability density, you first square the wave function

$$|\psi_2|^2 = \frac{2}{4.22 \text{ Å}} \sin^2\left(\frac{2\pi x}{4.22 \text{ Å}}\right) \quad \text{between } x = 0 \text{ Å and } x = 4.22 \text{ Å}$$

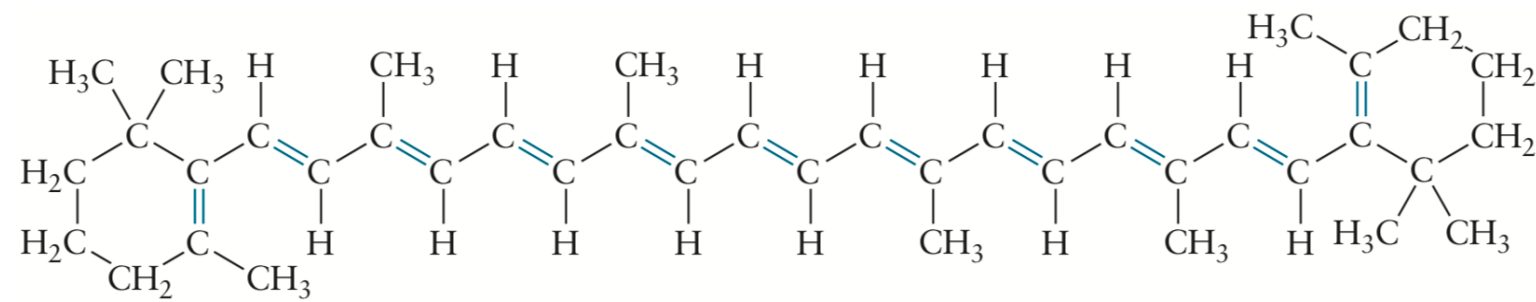
The maximum values of $|\psi_2|^2$ occur when the sin function is $+1$ and -1 , so at $2\pi x/(4.22 \text{ Å}) = \pi/2$ and $3\pi/2$. Thus the most likely positions of this $n = 2$ electron are $x = 1.055 \text{ Å}$ and $x = 3.165 \text{ Å}$. The minimum value of $|\psi_2|^2$ occurs at $x = 2.11 \text{ Å}$, so this is the least likely position of the $n = 2$ electron.

Electronic Spectroscopy

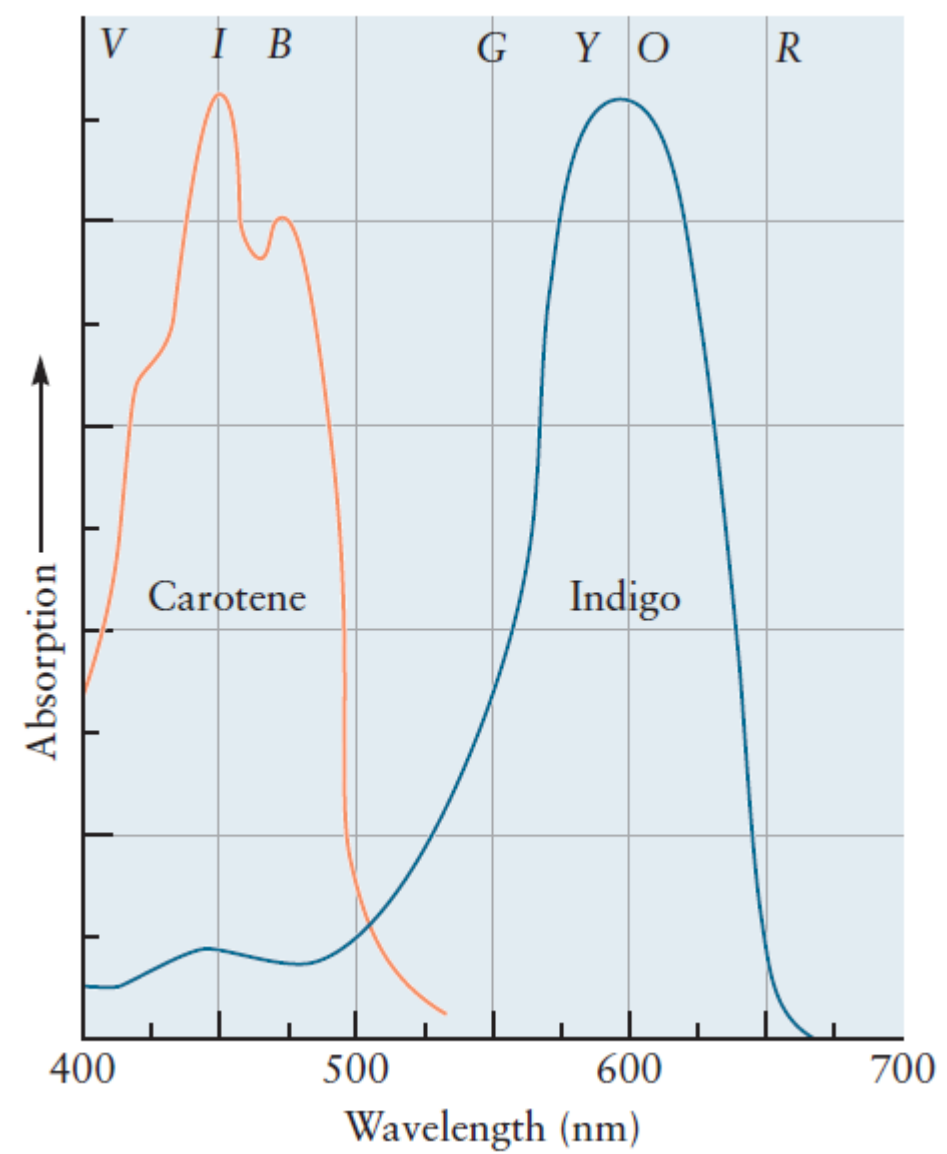
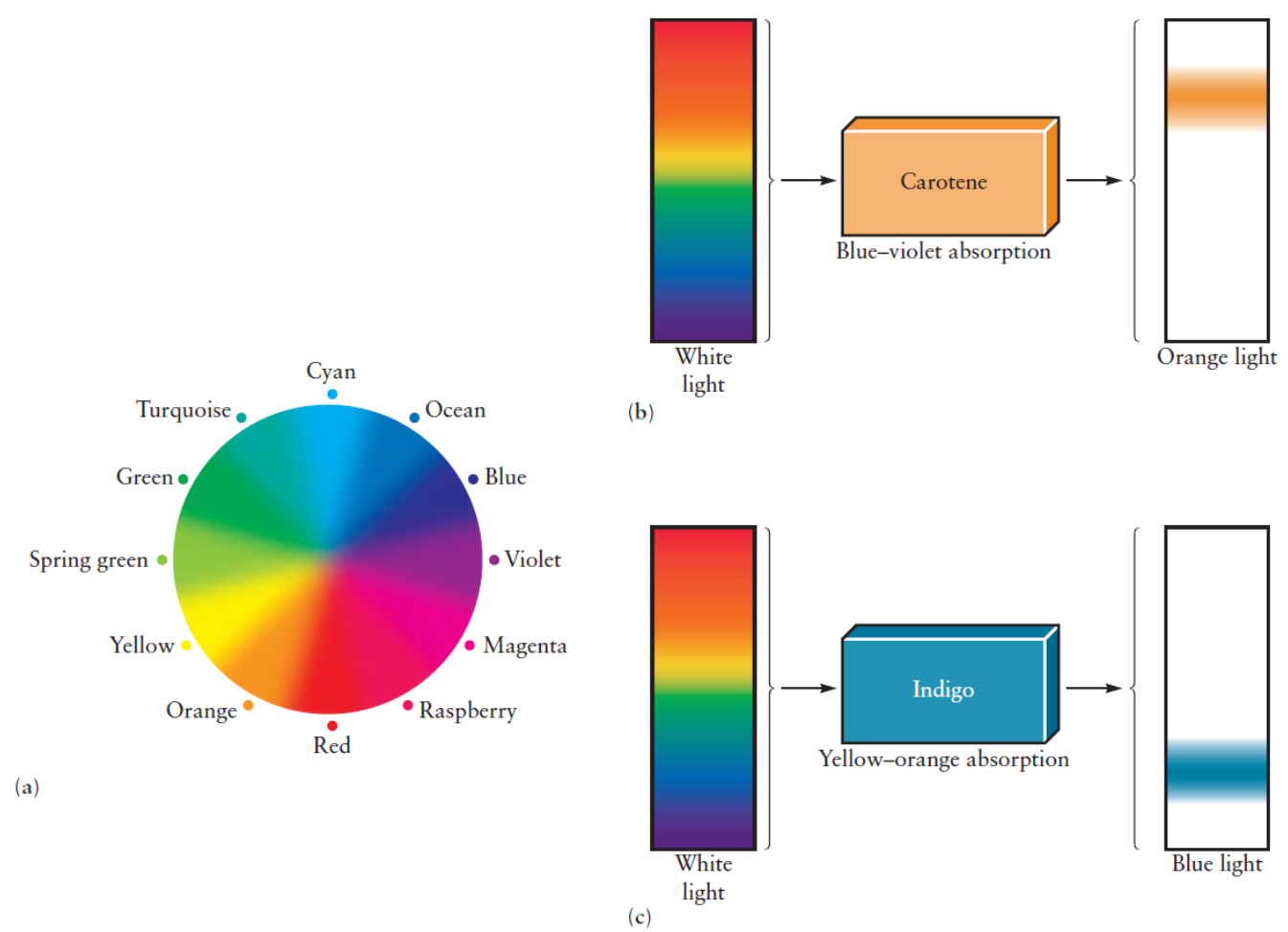
Now consider a molecule in which several double bonds alternate with single bonds in a conjugated π -bonding system like that for butadiene

TABLE 20.2 Absorption of Light by Molecules with Conjugated π Electron Systems

Molecule	Number of C=C Bonds	Wavelength of Maximum Absorption (nm)
C ₂ H ₄	1	162
C ₄ H ₆	2	217
C ₆ H ₈	3	251
C ₈ H ₁₀	4	304



Electronic Spectroscopy



Optical Properties and the Spectrochemical Series



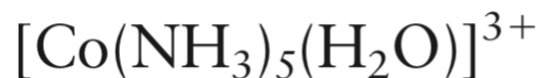
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Orange



A green form and a violet form



Purple

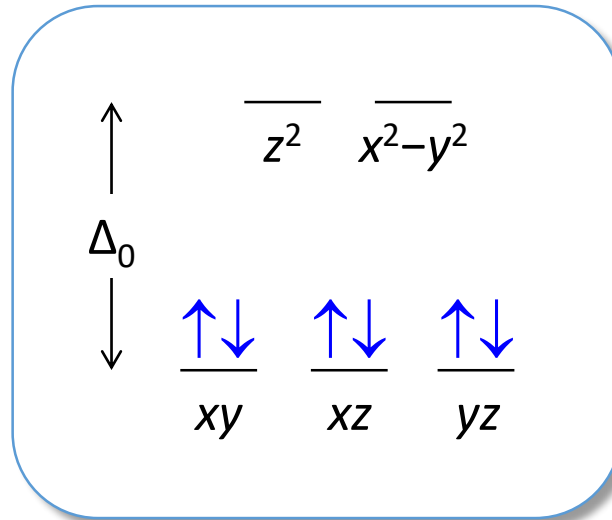
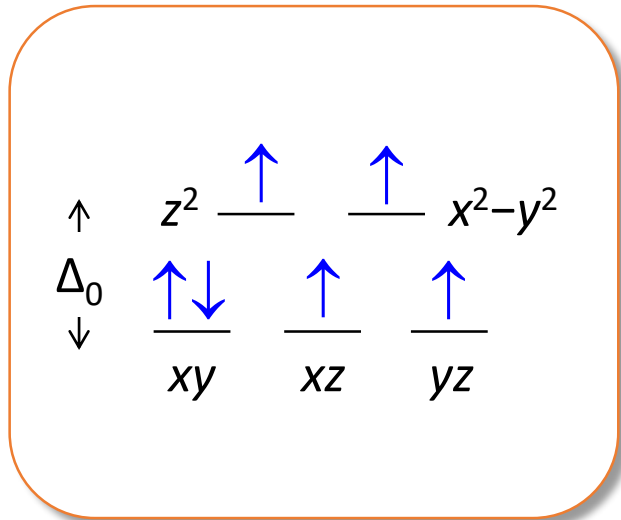
The colors depend on the

- **oxidation state of the metal ion**
- **number and nature of the ligands**
- **geometry of the complex**

FIGURE 8.23 Several colored coordination compounds. (clockwise from top left) They are $\text{Cr}(\text{CO})_6$ (white), $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ (green), $[\text{Co}(\text{en})_3]\text{I}_3$ (orange), $[\text{Co}(\text{NH}_3)_5(\text{OH}_2)]\text{Cl}_3$ (red), and $\text{K}_3[\text{Fe}(\text{CN})_6]$ (red-orange).

Optical Properties and the Spectrochemical Series

The wavelength of the strongest absorption is called λ_{max} and it is related to Δ_o

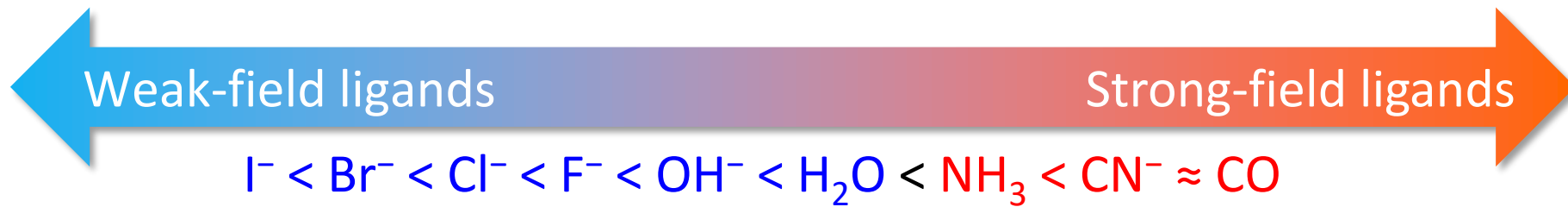


$$\Delta_o = h\nu = hc/\lambda_{\text{max}}$$

$[\text{Co}(\text{NH}_3)_6]^{3+}$	Orange
$[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$	A green form and a violet form
$[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$	Purple

High spin

Low spin



The smaller halide ions can approach the metal ion more closely: greater electrostatic repulsion.

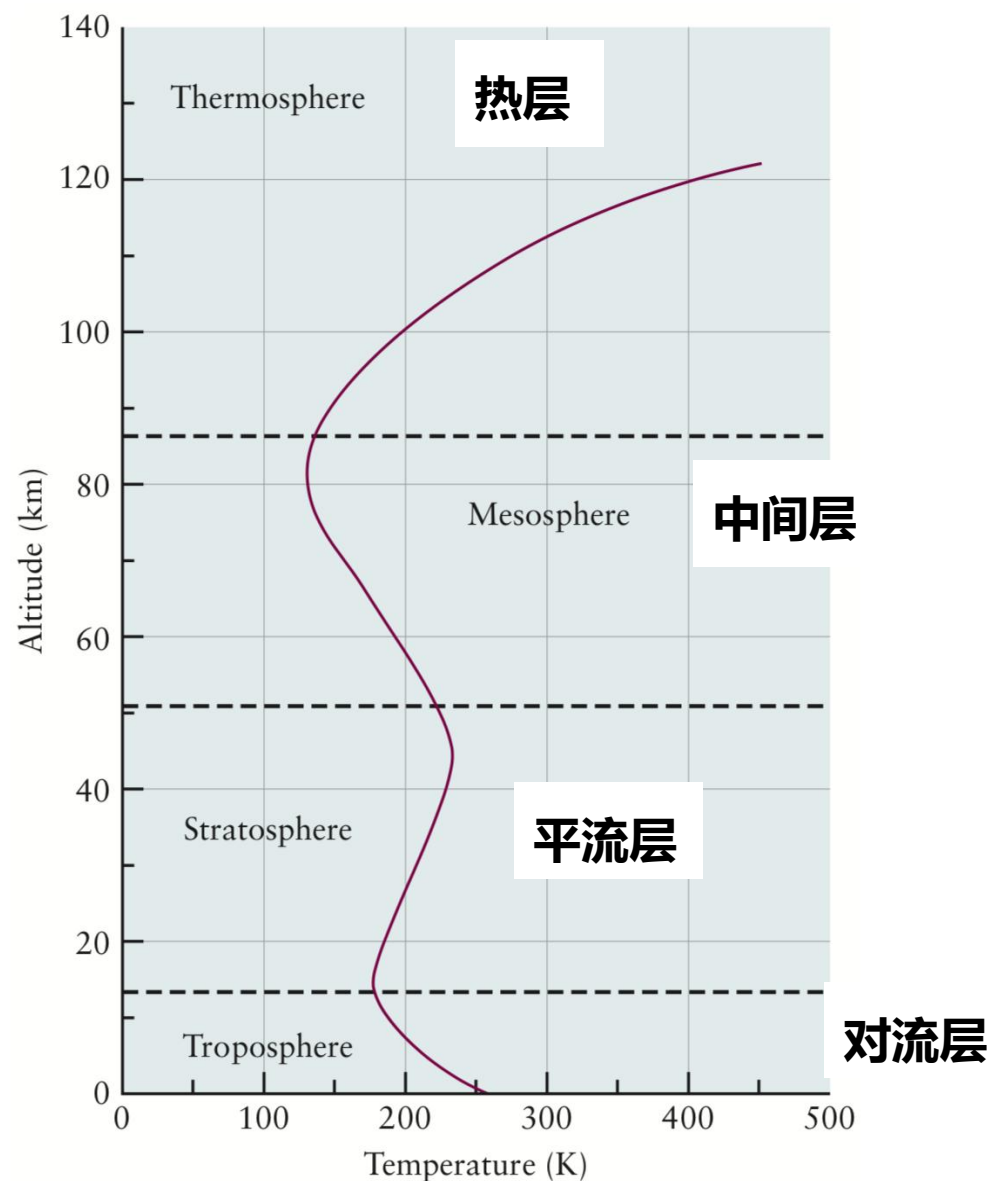
Suppose you set out to design a new green dye. Over what range of wavelengths would you want your trial compound to absorb light?

Major points of today's lecture

- **Molecular spectroscopy and photochemistry**

1. **Introduction to Molecular Spectroscopy**
2. **Experimental Methods in Molecular Spectroscopy**
3. **Rotational and Vibrational Spectroscopy**
4. **Nuclear Magnetic Resonance Spectroscopy**
5. **Electronic Spectroscopy and Excited State Relaxation Processes**
6. **Introduction to Atmospheric Chemistry**
7. **Photosynthesis**

Introduction to Atmospheric Chemistry



In the outer two layers (**the thermosphere and the mesosphere**): the density is low and the gases have been extensively ionized by intense, high-energy solar radiation.

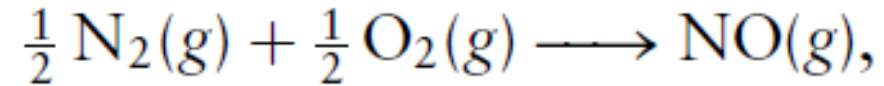
The third layer, **the stratosphere**, occupies the region from 12 km to 50 km (approximately) above the Earth's surface.

The troposphere is the lowest region, reaching up to about 12 km from the Earth's surface. In the troposphere, warmer air lies beneath cooler air.

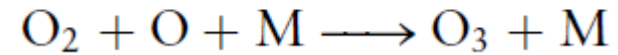
Tropospheric对流层 Chemistry

The **oxides of nitrogen** are major air pollutants.

- 10^{-10} ppm at room temperature
- 100 ppm at 800C

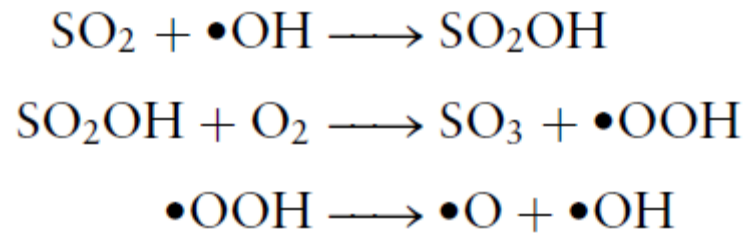


The photolysis of NO_2 is responsible for
the generation of ground-level ozone

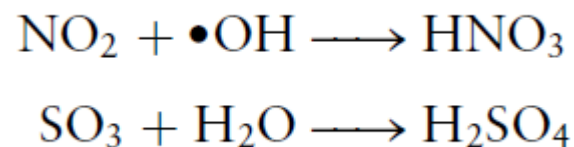


Tropospheric对流层 Chemistry

The oxides of sulfur create global pollution problems because they have longer lifetimes in the atmosphere than the oxides of nitrogen



Acid rain results from the reaction of NO_2 and SO_3 with hydroxyl radical and water vapor in the air to form nitric acid (HNO_3) and sulfuric acid (H_2SO_4). The acids dissolve in water and return to the earth in the rain.



Tropospheric对流层 Chemistry

T A B L E 20.10	
National Ambient Air Quality Standards (Primary) [†]	
Pollutant	Level
Carbon Monoxide	
8 hour average	9 ppm (10 mg/m ³)
1 hour average	35 ppm (40 mg/m ³)
Lead	
Rolling 3 month average	0.15 μg/m ³
Quarterly average	1.5 μg/m ³
Nitrogen dioxide	
Annual average	53 ppb
1 hour average	100 ppb
Particulate matter (PM₁₀)	
24 hour average	150 μg/m ³
Particulate matter (PM_{2.5})	
Annual average	15.0 μg/m ³
24 hour average	35.0 μg/m ³
Ozone	
8 hour average	0.075 ppm (2008 std.)
1 hour average	0.12 ppm (some regions)
Sulfur dioxide	
Annual average	0.03 ppm
24 hour average	0.14 ppm
1 hour average	75 ppb

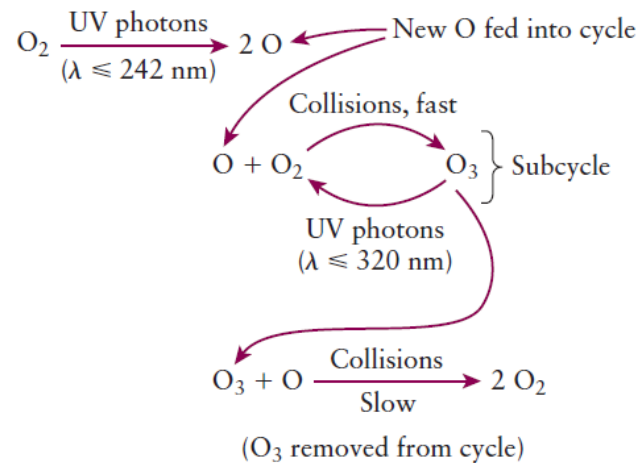
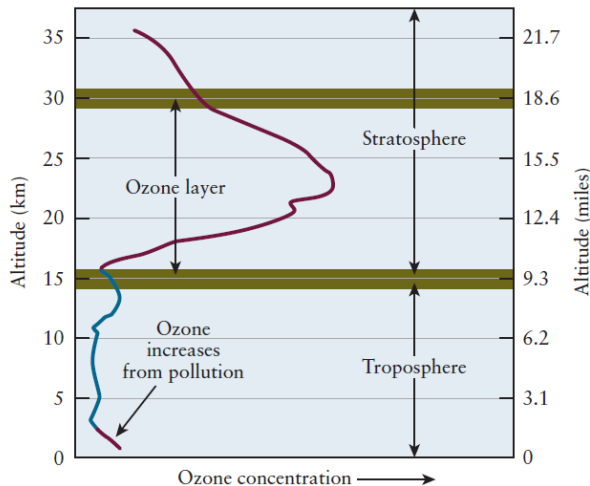
[†]These standards are updated periodically. Please check the U.S. Environmental Protection Agency website (<http://epa.gov/air/criteria.html>) for the current standards.



Stratospheric 平流层 Chemistry

Sun: resembles a 6000K black body which generates a considerable number of very high-energy UV photons strikes the Earth's atmosphere. The energy of a 200 nm photon is more than 6 eV, which is sufficient to break all but the strongest chemical bonds.

A substantial layer of ozone in the stratosphere that absorbs most of this harmful ultraviolet radiation, affording us substantial protection.

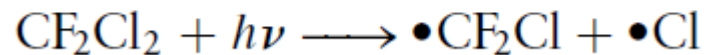
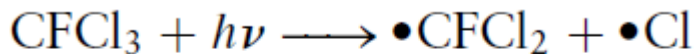


This reaction reduces the number of high-energy photons reaching the lower parts of the atmosphere, especially those with wavelengths less than 200 nm.

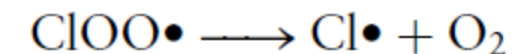
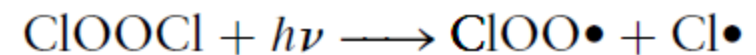
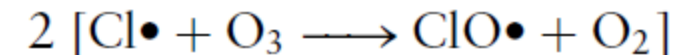
Stratospheric 平流层 Chemistry

One pressing concern involves chlorofluorocarbons (CFCs)—compounds of chlorine, fluorine, and carbon used as refrigerants and as propellants in some aerosol sprays. CFCs are nonreactive at sea level but can photodissociate in the stratosphere.

Photolysis of CFCs with <200 nm wavelength UV light produces chlorine atoms and radical fragments



The chlorine atom is a reactive free radical destroys ozone by the catalytic mechanism



Stratospheric 平流层 Chemistry

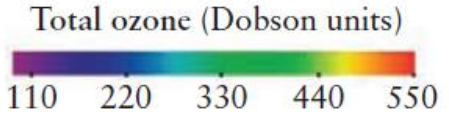
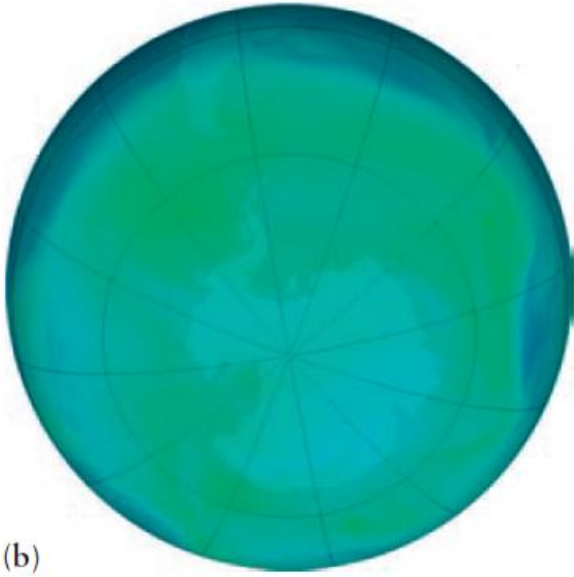
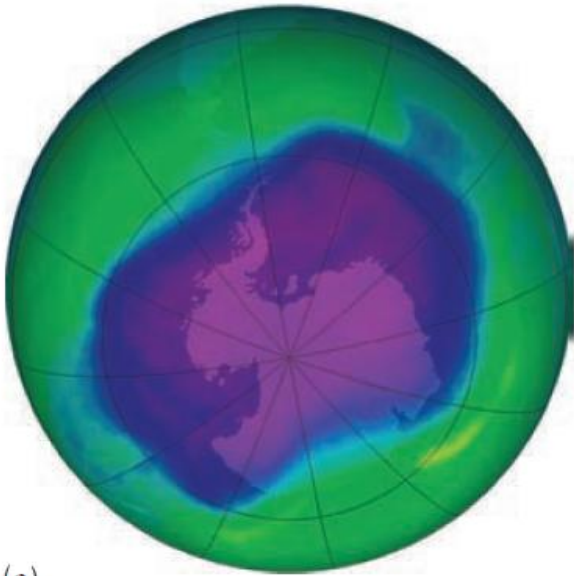
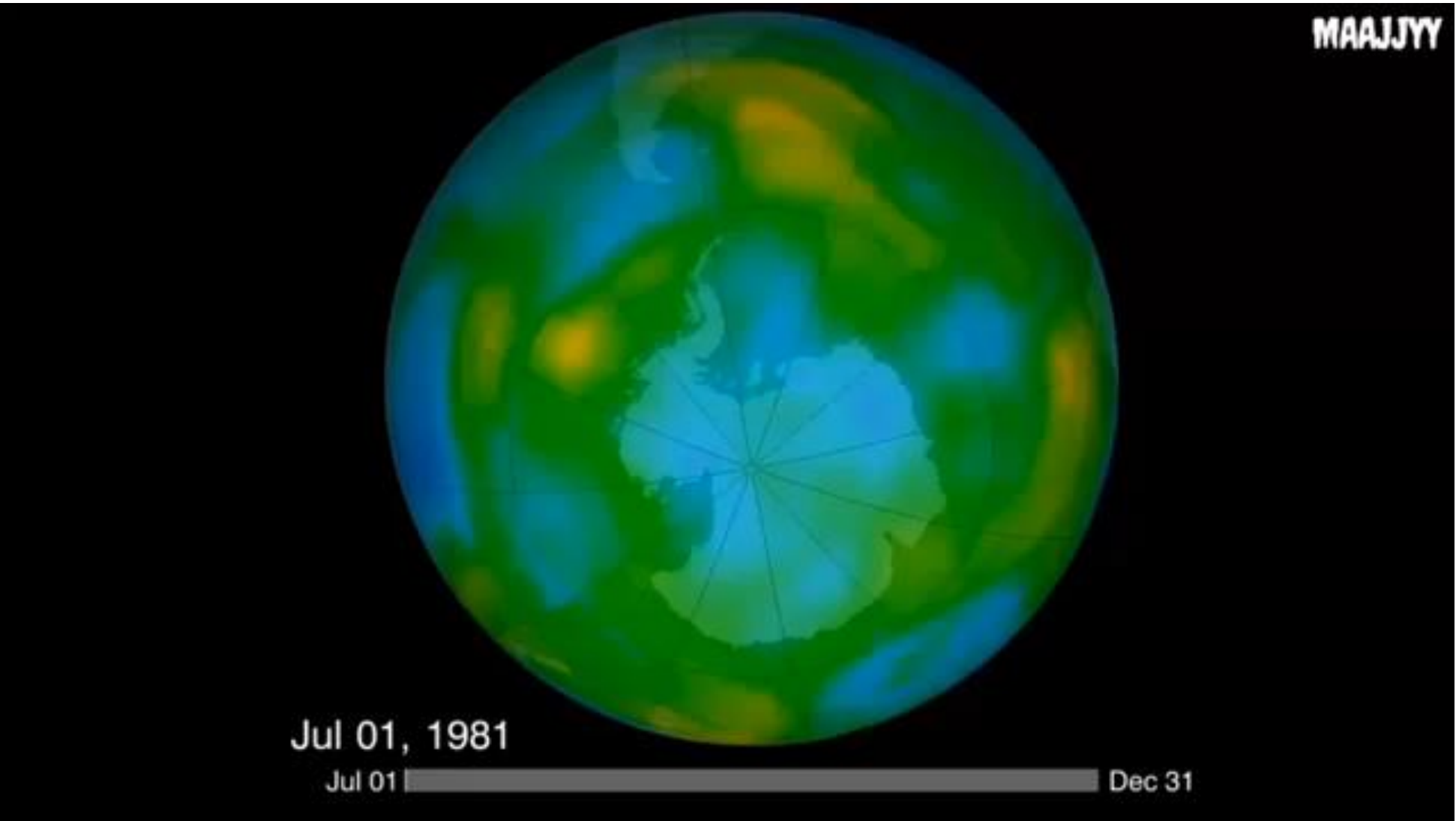




Photo from the Nobel
Foundation archive.

Paul J. Crutzen

Prize share: 1/3

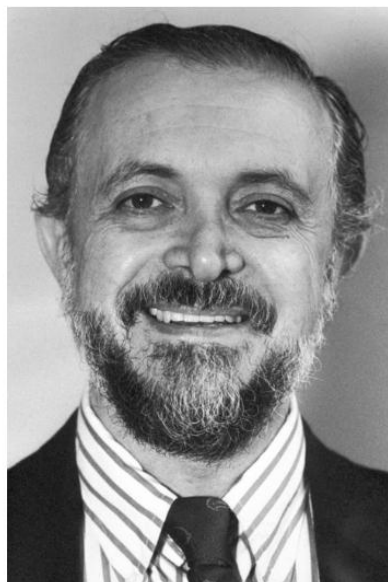


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Foundation archive.

Mario J. Molina

Prize share: 1/3



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F. Sherwood Rowland

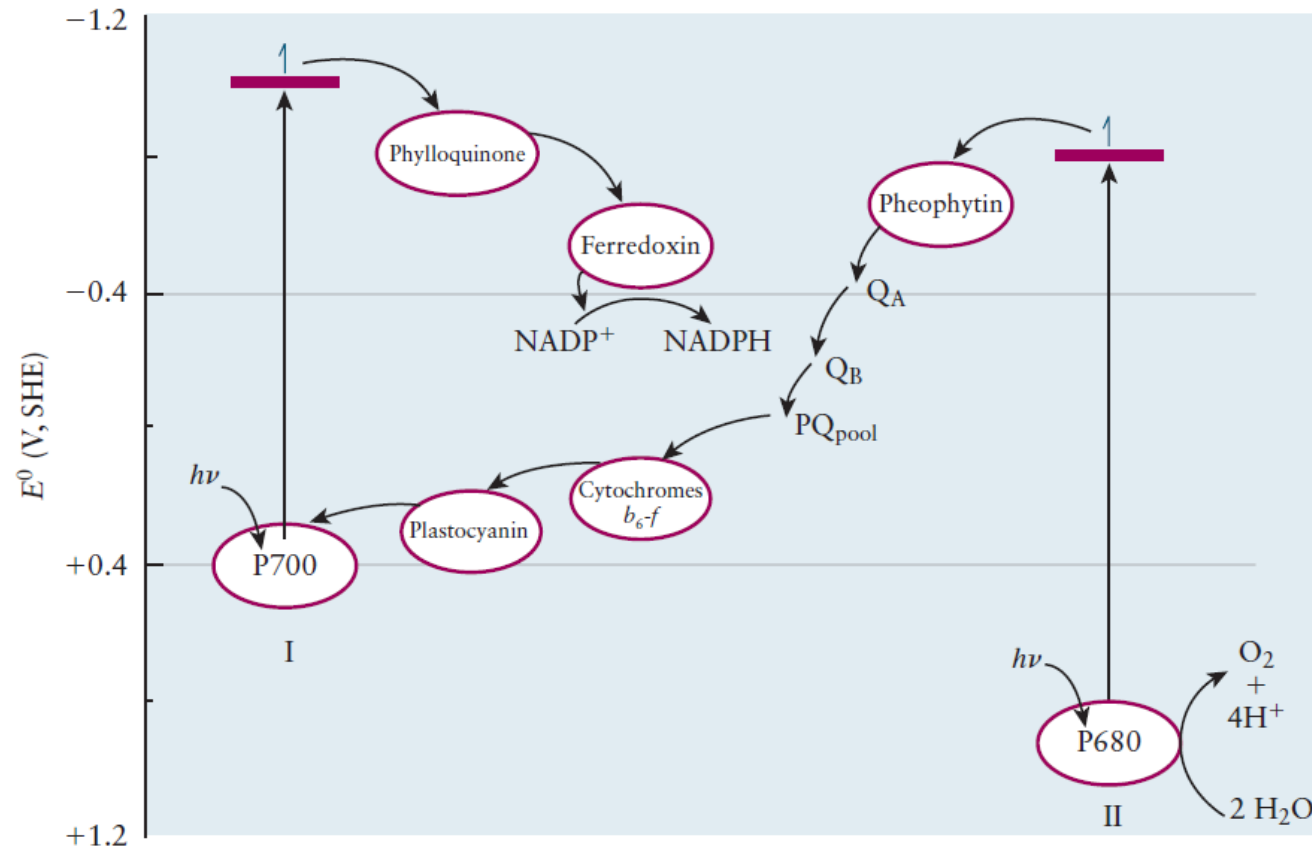
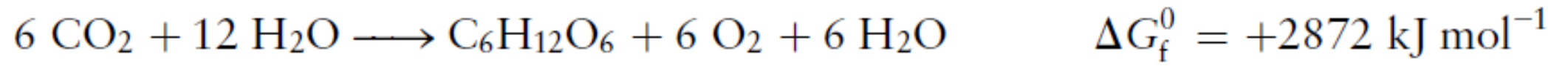
Prize share: 1/3

The Nobel Prize in Chemistry 1995 was
awarded jointly to Paul J. Crutzen, Mario J.
Molina and F. Sherwood Rowland "for their
work in atmospheric chemistry, particularly
concerning the formation and decomposition
of ozone"

<https://www.nobelprize.org/prizes/chemistry/1995/summary/>

Photosynthesis

photosynthesis—a remarkably efficient sequence of photo-physical processes and oxidation–reduction reactions that leads to the formation of a six-carbon sugar molecule plus molecular oxygen as follows



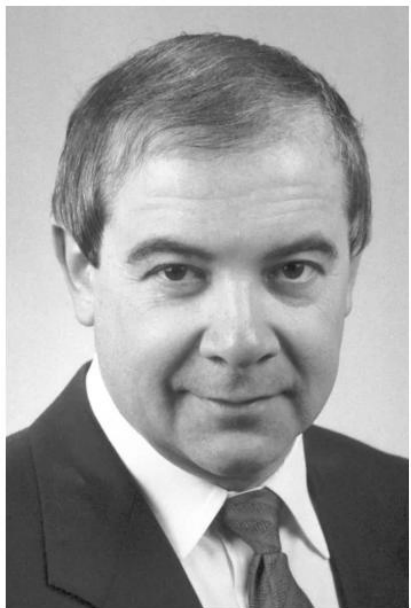


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Foundation archive.

Johann Deisenhofer

Prize share: 1/3



Photo from the Nobel
Foundation archive.

Robert Huber

Prize share: 1/3



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Foundation archive.

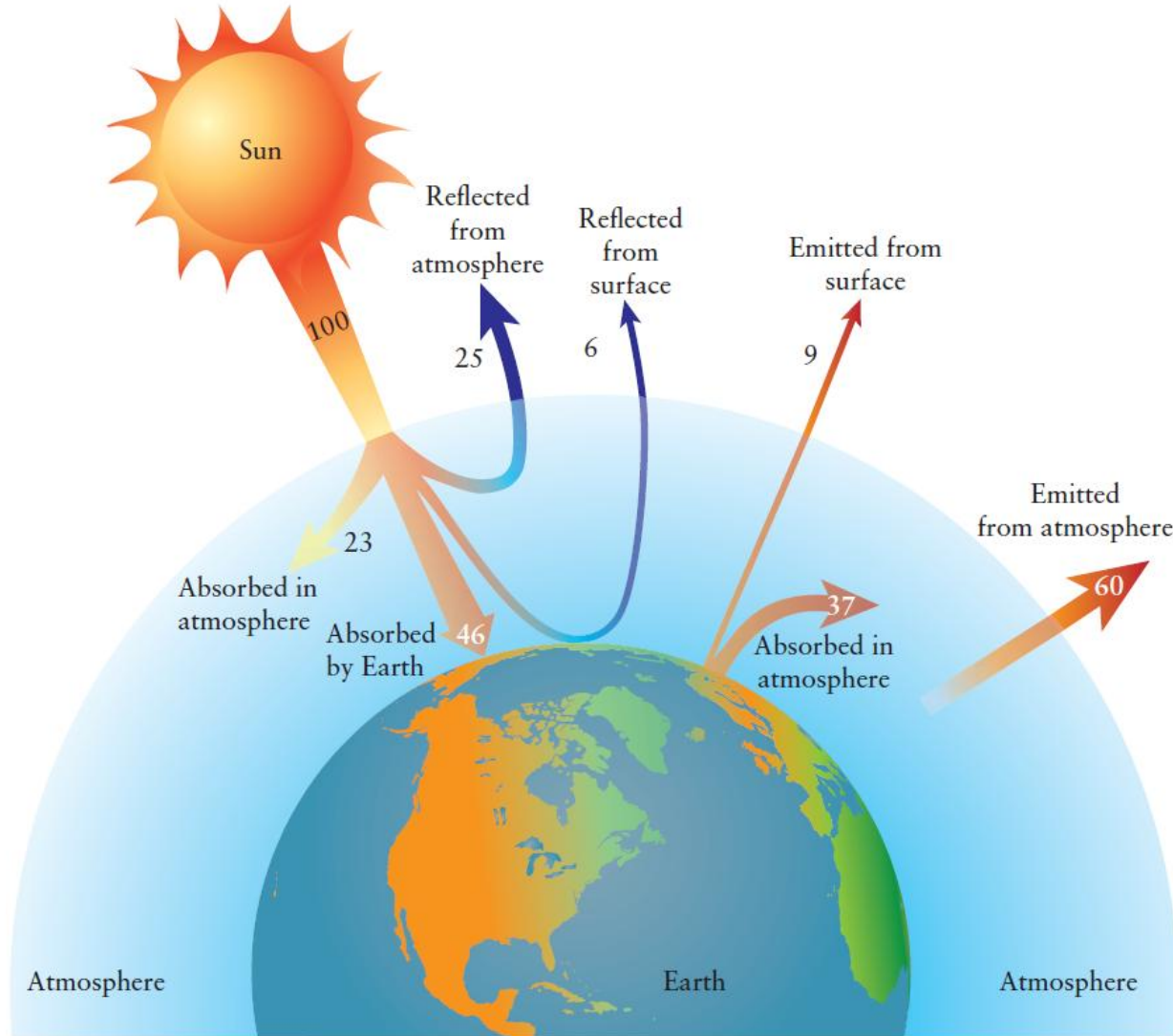
Hartmut Michel

Prize share: 1/3

The Nobel Prize in Chemistry 1988 was
awarded jointly to Johann Deisenhofer,
Robert Huber and Hartmut Michel "for the
determination of the three-dimensional
structure of a photosynthetic reaction
centre"

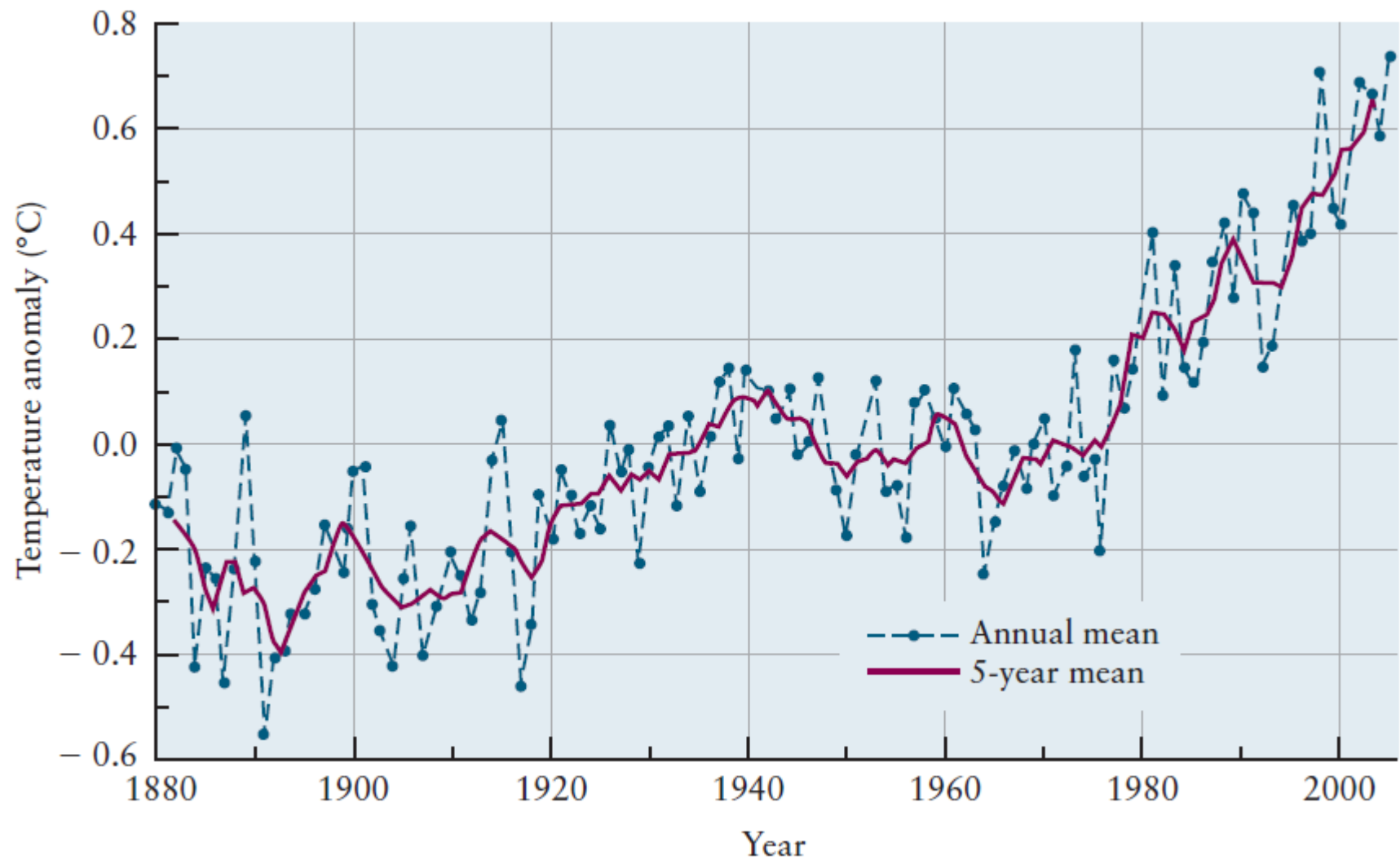
<https://www.nobelprize.org/prizes/chemistry/1988/summary/>

Climate Change: The Greenhouse Effect

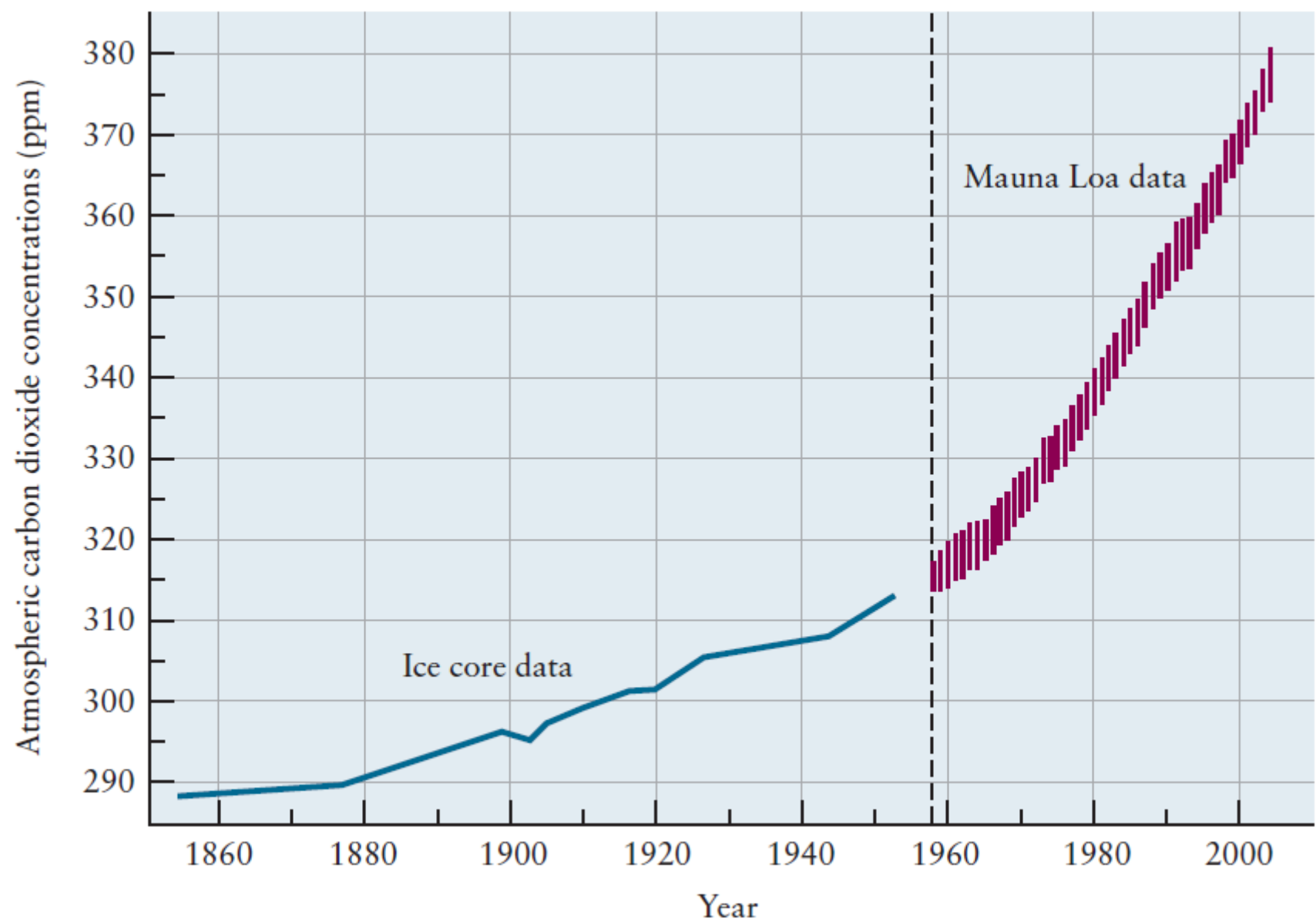


Certain gases in the troposphere 对流层 play a crucial role in this balance because they absorb infrared radiation emitted by the warm surface of the earth rather than letting it pass out to space

Climate Change: The Greenhouse Effect



Climate Change: The Greenhouse Effect



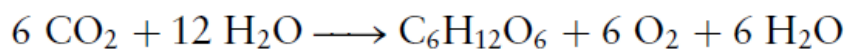
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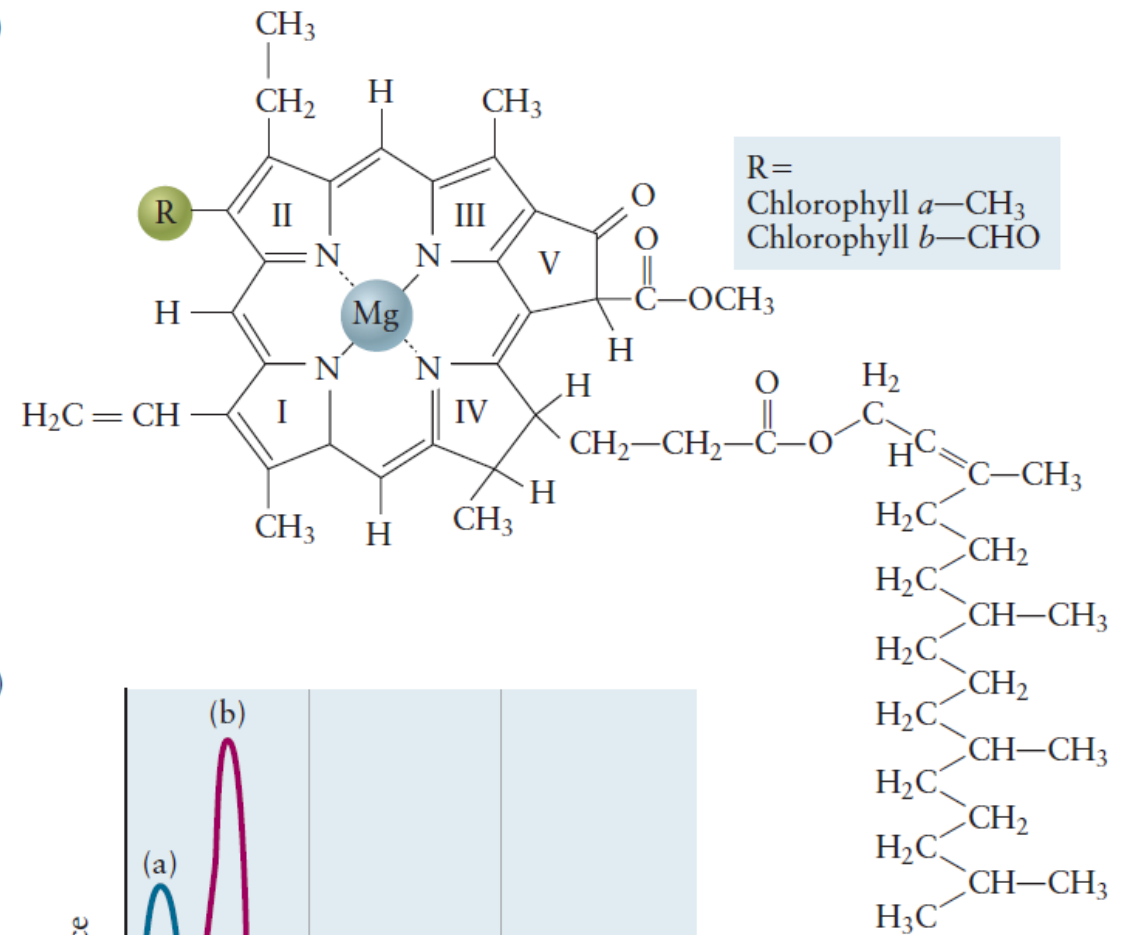
Photosynthesis

~1% of the energy that reaches the Earth's surface is converted into chemical energy by photosynthesis: a photophysical processes and oxidation–reduction reactions that leads to the formation of a six-carbon sugar + oxygen

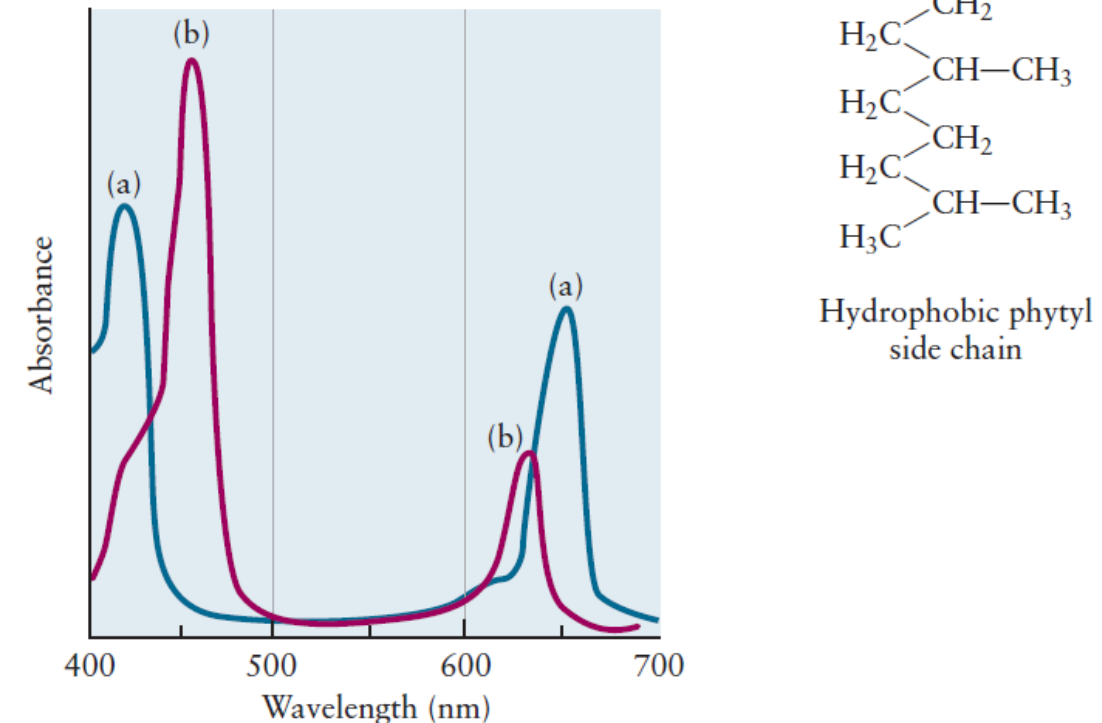


$$\Delta G_f^0 = +2872 \text{ kJ n}$$

(a)



(b)



Homework: read “principle of modern chemistry”, chapter 20